

Name: Bol
Datum: 29.07.2004

Letzte Position: 29

INTEGRAL HYDRAULIK
Anlagen + Systeme

Geräteliste zu Schaltplan: 46.080 HSMS 850/10-6/2

Narayangonj

Pos.	Stck.	ID-Nr.	Zeichnung-Nr.	Benennung	Typenbezeichnung		Lieferant
1.	1	11934	859-1060-010-048	Geräteschrank / hydraulik cabinet	V=850 l		Integral Hydraulik
2.1	2	12015	110-1323-016-270	Zahnradpumpe / external gear pump	MLP D/G 211 C	11 ccm/U	Integral Hydraulik
	2	12242	200-1061-084-027	Pumpenträger / bell housing	PL 200/02/03		KTR
	2	12243	242-2295-824-002	Kupplung / coupling	R 24.24-N/2A		KTR
2.2	2	12013	060-1323-016-270	Zahnradpumpe / external gear pump	MLP D/G 206 C	6 ccm/U	Integral Hydraulik
	2	12242	200-1061-084-027	Pumpenträger / bell housing	PL 200/02/03		KTR
	2	12243	242-2295-824-002	Kupplung / coupling	R 24.24-N/2A		KTR
3.1	2	12244	001-2442-009-034	E-Motor / electric motor	1,1 KW;B3/B5;380V;50Hz;1000min-1;IP54;ISO F/B		Siemens
3.2	2	12244	001-2442-009-034	E-Motor / electric motor	1,1 KW;B3/B5;380V;50Hz;1000min-1;IP54;ISO F/B		Siemens
4.1	2	11976	001-1321-009-012	Manometer / pressure gauge	MG100/-1...		Integral Hydraulik
4.2	2	12037	100-2417-025-010	Manometer / pressure gauge	MG100/10B		Integral Hydraulik
5.1	1	12216	001-1332-009-034	Kugelhahn / ball valve	Best. Nr.: 90 025 04 3/4"		AS
	1	11090	034-2208-000-000	Verschlussschraube / screw connector	DIN 906-G3/4 /4-PH		
5.2	2	11973	001-1332-009-019	Kugelhahn / ball valve	BHG V400.312-1		Böhmer
5.3	2	11973	001-1332-009-019	Kugelhahn / ball valve	BHG V400.312-1		Böhmer
5.4	2	12245	041-2453-400-016	Minimessanschluss / gauge port			Hydrotechn.
5.5	2	12246	040-2453-400-022	Minimessanschluss / gauge port			Hydrotechn.
5.6	2	12246	040-2453-400-022	Minimessanschluss / gauge port			Hydrotechn.
6	2	12247	001-1330-009-058	Druckbegrenzungsventil / pressure relief valve	SPV10 A1G1 A12		Kracht
9	2	12249	016-1531-014-014	Rückschlagventil / check valve	RHD16-PSR-ED nach LOESCHE-Datenblatt		Integral Hydraulik
18	2	12189	022-1334-024-022	Drosselventil / flow restrictor	FT257/2-3/4		LT
19	1	12056	001-1327-009-456	Belüftungsfilter / air breather	Pi 0145 Mic		Mahle
20	2	11921	001-1327-009-083	Niederdruckfilter / filter unit	Pi 2008-57, 766.521.9		Mahle
	2	11876	001-1327-009-493	Filterelement / filter element	Pi 4108 Sm25		Mahle
21	1	11989	001-1304-009-071	Thermometer / temperatur	8201-20-10		Jumo

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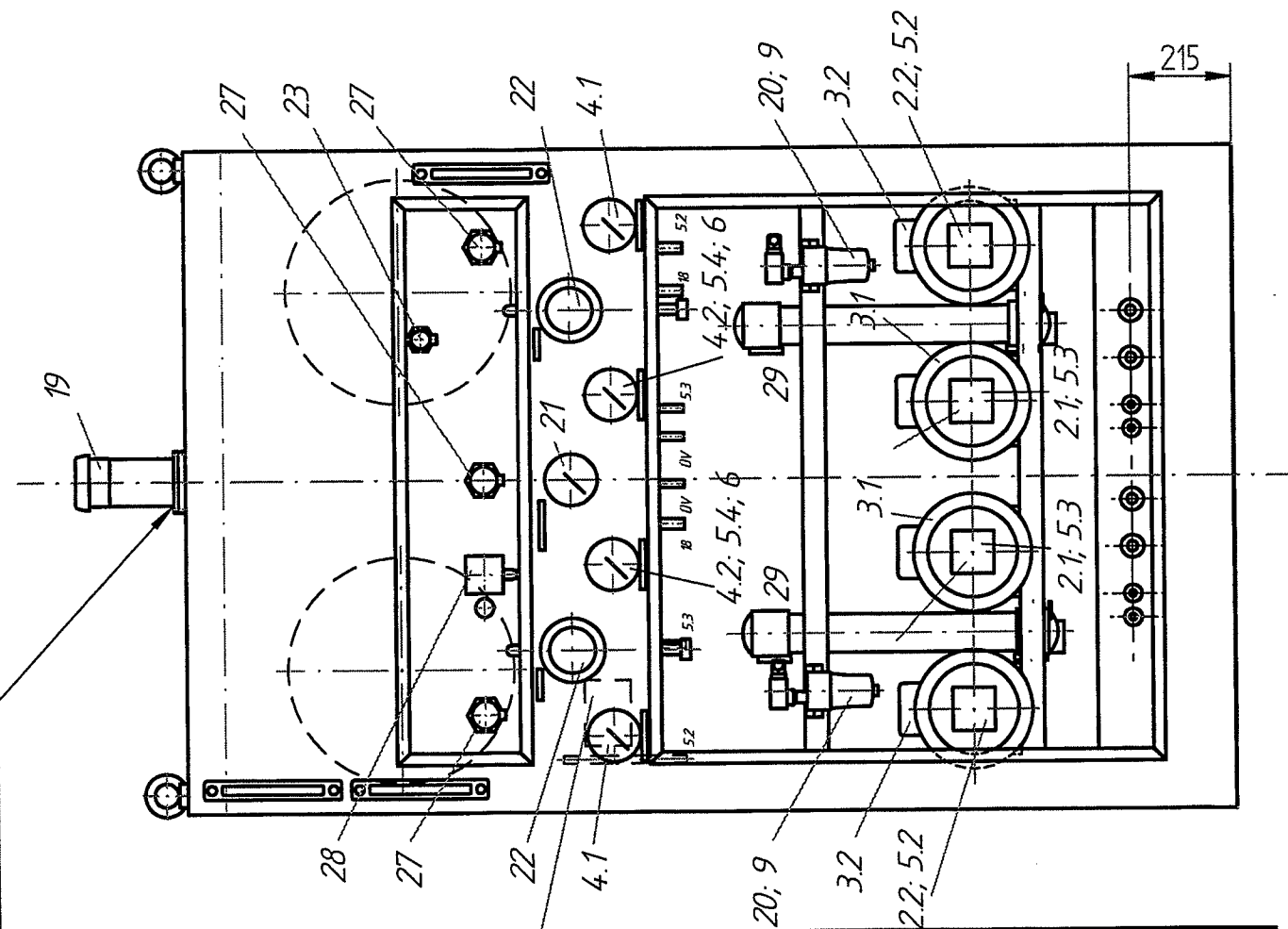
INTEGRAL HYDRAULIK
Anlagen + Systeme

Geräteliste zu Schaltplan: **46.080 HSMS 850/10-6/2**

Narayangonj

Pos.	Stck.	ID-Nr.	Zeichnung-Nr.	Benennung	Typenbezeichnung	Lieferant
22	2	12250	001-1360-009-008	Schauglas / observation window	Sk84	Thum
	1	12197	313-2487-000-000	Ölstandsanzeige / oil level gauge	UC-FI-313	UCC
23	1	12251	001-1305-009-110	Niveauüberwachung / zero overlap	SSM.1.B4.100.4.S1	Engler
26	1			Klemmenkasten / terminal box		Rose
27	3	12155	001-1305-009-385	Heizstab / heating element	G2; 800mm lang; 380V;900 W; Nr.: 32601402	Helios
28	1	16577	001-1305-009-568	Widerstandsthermometer / electric resistance thermometer	WBV10; R1/2"; Pt100 Klasse A	Grillo
29	2	12253	001-1304-009-064	Wärmetauscher / heat exchanger	T-8061-A-4-OTT	Flovex

Vor dem Einstecken des Belüftungs-
rohres O-Ring 98x1,5 (ZN 098-2250-000-015)
einlegen.



INTEGRAL HYDRAULIK Typ

Zeichen Änderungen Ä.-M.-Nr Datum Name

Datum	Name	zeichnung Nr.
gez. 29.07.04	Bol	046-1462-013-080
gepr. "		
N. gepr. "		

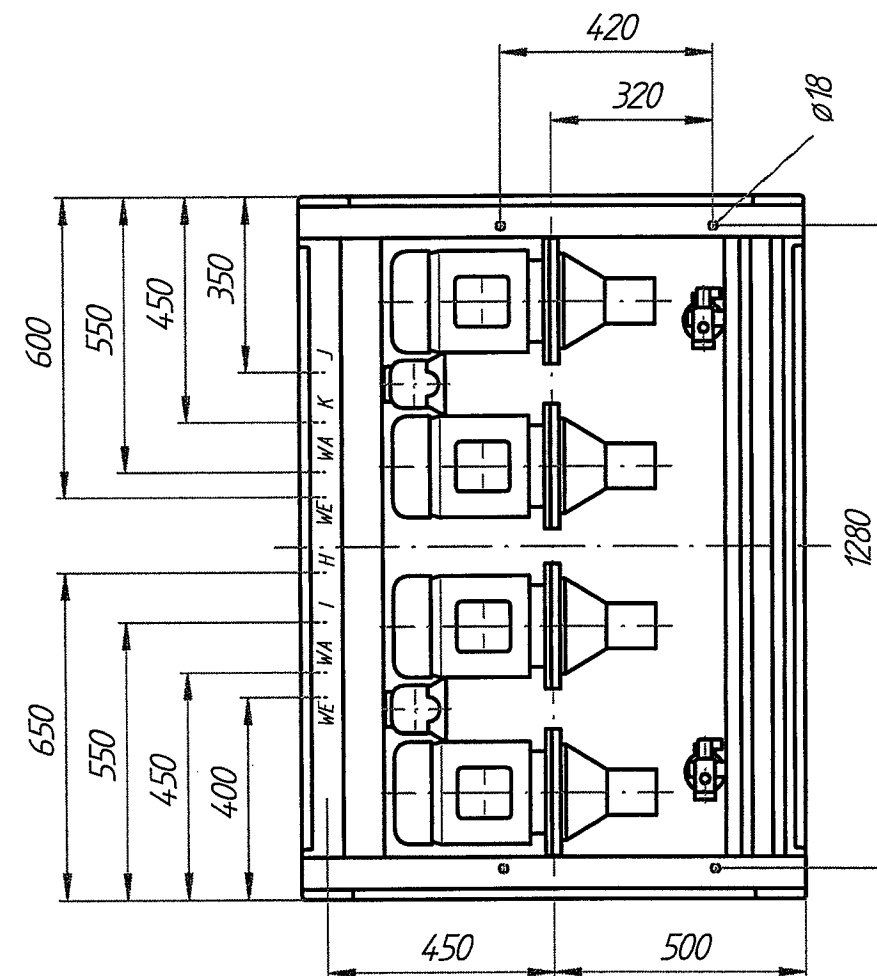
INTEGRAL HYDRAULIK

Anlagen + Systeme

*Ohne Verkleidungen gezeichnet.
Drawing without cover.*

*WE/WA = Wasser-Rohr G1/2
Außengewinde*

SV28-I
7-82AS



Ges.-Gew.		620 kg	Betreff:		Narayanganj						LOESCHE			
Gez.			Ersatz fuer											
Gepr.			Ersetzt durch											
Gez.			File-Nr. :											
Gepr.														
Oberflaechenzeichen Mittenrauhwert R in um Masse ohne Toleranzangaben nach DIN 7168 Guetegrad mittel			✓ Antiefer- zustand	✓ Ra < 50	✓ Ra < 12,5	✓ Ra < 1,6	✓ Ra < 0,4	✓ Ra < 0,1	CAD		EDV			
Maßstab		Hydraulikschrank								G46.080				
1:10		HSMS 850/10-6/2												
		Hydraulik Cabinet												
Urheberschutz nach DIN 34 vorbehalten												Total	RI	RI.Nr.

Installation instructions Start-up, upkeep and maintenance General hazard information

1. PIPE-LINES

1.1 GENERAL

For the installation and erection of hydraulic lines, the following are necessary: hand tools, bending rig, grinder and as well as the possibility to store erection materials.

Discussions to reach agreement between the Client and the Contractor with respect to the pipe routing are also a part of the preparation.

The installation of hydraulic piping should take place in clean surroundings in order to avoid contamination of the hydraulic system, for which the working area must be protected against dust and other contaminants.

During installation, connections to the piping and to the equipment and cylinders must remain closed. Before making connections, connectors and the corresponding connecting surfaces are to be checked for cleanliness.

The installation of the hydraulic piping network must be carried out carefully, whereby special attention is to be given to:

- Cleanliness of the work
- Straight, direct and easy-to-understand pipe routing
- Installation of the piping without tension
- During the installation, attention must be paid to
- ensure no dirt gets into the hydraulic system
- A straight-line pipe routing is to be preferred to make maintenance and repairs much easier
- Piping is to be relocated without tension

After the pipe routing has been established, the work at the erection site can begin with the installation of the pipe clamps necessary for the pipe routing. The spacing between two pipe clamps (restraints) should amount to 1 to 1,5 m. It is very important that the piping is installed without tension.

To make the installation as well as later maintenance and repair easier, the locations of connections should be offset with the necessary spacing to the next pipe.

Contacts between the pipe and construction elements should be avoided, especially with the installation of stainless pipes.

The pipe is best cut to length with the help of an appropriate machine. Should it be necessary to saw a pipe to length, it is absolutely necessary, that the pipe be deburred and leaned. This can be done using available compressed air, with which the pipe is blown out. The bending of pipes to be preferred to the use of threaded fittings. Bending is faster, cleaner and also cheaper.

Threaded angle fittings are connections which can work loose and result in leaks.

1.2 INSTRUCTIONS FOR WELDED PIPES AND CONNECTIONS

When welding on the threaded fittings foreseen for the hydraulic system, a weld nipple is welded to the pipe. The method of welding corresponds to the usual technical possibilities today, whereby there are no problems with leak-tightness between the pipe and the weld nipple.

O-rings are used to provide a sealing between the weld nipple and the threaded fittings, so that no problems are to be expected here with proper work. The holding function between the weld nipple and threaded fitting is made with the help of a threaded collar.

The welding process to be recommended is the TIG process (tungsten inert gas), whereby the electrical current is sent through a fixed tungsten electrode surrounded by Argon gas. The filler material is introduced by hand or automatically, whereby a forming gas is introduced through the pipe in order to avoid burning through the wall into the inside. Because of the use of both gases, the TIG welding process is very clean, meaning that no slag is formed. Nevertheless, the welding seams should be pickled after welding in order to remove slag and any welding beads from the pipe.

After being pickled, the pipe must be passivated.

After the pipe installation is completed, the individual lines are flushed. The use of a separate flushing unit is to be preferred.

One can assume that flushing will require approx. 1 hour per pipe. The flushing medium must be compatible with the operating medium. The flushing of the pipe is made with the cylinder and hydraulic equipment clamped off. When the flushing is completed, the piping is connected to the hydraulic system. All connections that can be loosened must be tested for tightness, meaning that all threaded collars must be tightened by hand also using an appropriate wrench.

The system is ready for operation when this work is concluded.

1.3 ASSEMBLY INSTRUCTIONS FOR PROGRESSIVE RING FITTINGS

1.3.1 General

The basic feature of the progressive ring is that precisely designed geometry permits the limited, progressive penetration of the ring into the tube. The front cutting edge has already begun to cut into the tube before the second cutting edge contacts the tube. When the optimum prescribed depth for both cutting edges has been reached, the top edge prevents further penetration. Because of the stop edge and the shape of the two cutting edges, the forces arising are distributed along the entire taper length, thus increasing joint strength. This distribution of the forces and the specially designed internal contour at the collar end of the ring guarantee increased reliability for flexure loading. Vibrations are thereby dampened at the entry into the ring. During assembly, the stop edge causes the tightening forces to be increased up to a clearly perceptible fixed point. (assembly fixed point). After assembly is completed, the visible thrown-up collar must completely fill out the space in front of the first cutting edge.

1.3.2 Preparation of the fitting components

Oil thread and taper of the fitting as well as ring and thread of the nut. Slide nut and ring onto the end of the tube. Do not slide ring on backwards. Tighten nut by hand into tight.

Always press the tube against stop in inner cone of the fitting body. Mark a line on nut and tube in order to observe the required number of turns. Tighten the nut by approximately 1½ turns. (tube must not turn with nut). Stop edge limits tightening by increasing force required.

1.3.3 Inspection of assembly

Loosen nut, check to make sure that a visible collar fills out the space in front of the first cutting edge (it does not matter if the ring can be rotated on the tube end). All connections that can be loosened must be tested for tightness, meaning that all threaded collars must be tightened by hand also using an appropriate wrench. The system is ready for operation when this work is concluded.

1.3.4 Re-assembly

For every re-assembly the nut must be retightened without applying more force than for the first assembly. The fitting body must be firmly held during all assemblies.

1.4 FLUSHING OF THE SYSTEM

After the pipe installation is completed, the individual lines are flushed. The use of a separate flushing unit is to be preferred. One can assume that flushing will require approx. 1 hour per pipe. The flushing medium must be compatible with the operating medium. The flushing of the pipe is made with the cylinder and hydraulic equipment clamped off. When the flushing is completed, the piping is connected to the hydraulic system.

2. PROVISIONS FOR START-UP PROCEDURES

2.1 PIPELINES

The connections between the pipelines of the equipment and the machines (at the consumers) must be made by trained hydraulic experts. The pipes and fittings used must comply with the pressure rating the equipment is designed for. All the pipelines must be held by exactly fitting mounting brackets. The pipes must be seamless hydraulic pipes pursuant to DIN 2391/C, St. 35.4, NBK and bonderized. Corresponding hydraulic pipes made of high-grade steel may also be used.

Tubing must be effected as straight and short as possible. Use of angles and knees is to be avoided. The number of screwed joints must be kept as low as possible, because every screwed connection represents a leakage risk. The dimensions of all connecting pipes must comply with the specifications given in the circuit diagram. The use of pipe with too narrow an inside diameter or with a length longer than stipulated leads to pressure losses that may negatively affect the performance of the equipment.

2.2 CLEANLINESS

Without exception all the pipes used for the oil-hydraulic equipment must be clean inside and free of all rust or dirt particles.

After installation they may not be welded or soldered any more. All pipes must be installed carefully, exactly, and free of stress.

Sharp edges and burrs at the end of the pipes are to be removed.

If threads at fittings are sealed with liquid sealing agents, care shall be taken that the nose of the thread remains free, otherwise hardened particles of the sealing may find their way into the system.

Remove any vent plugs in the connections of the equipment only directly prior to connecting the relevant lines. This helps to keep the equipment clean.

2.3 MOTORS AND ACCUMULATORS

When installing the electric motor, it is indispensable to take care of the alignment between pump and motor shaft. This applies in particular to type B 3.

However, even in the case of complete, ready-assembled equipments of this type the equipment may suffer from distortion caused by transport. Therefore it should always be checked before start-up.

The values for voltage and amperage stated on the electric motors and components must correspond with the factual energy supply. After connecting the electric motor, the sense of rotation is to be checked. This is done by a short start. The sense of rotation is shown by an arrow on the pump. If the pump runs in the wrong direction for more than a few seconds, it will lead to serious damage caused by dry run.

Gas-filled accumulators must be brought to the correct refill pressure. The position of the deaerating and shut-off valves is to be checked according to the circuit diagram. Unless otherwise stated, nitrogen is used as a gas medium. Manufacturer's provisions must unconditionally be complied with when filling, mounting, and maintaining the accumulators.

2.4 HYDRAULIC FLUID

The hydraulic fluid is to be handled very carefully. Particular attention shall be paid to the use of the proper fluid and to the fact that it is absolutely clean. Special care shall be used if the oil is taken from a barrel. It shall be checked whether the contents are free of any contamination, such as rust or condensing water.

The medium shall be filled through the filler neck. If possible a pump-over unit with a fine filter is used for filling.

All parts shall be thoroughly scavenged before the equipment is run under full speed and working pressure. If at all possible, the same fluid is used for this purpose that is later used for normal operation. The consumption connections are bridged by hose connections and very sensitive devices are replaced by scavenging plates during the scavenging. If the pump belonging to the equipment has to be used for scavenging, care has to be taken that it only delivers clean fluid and working pressure is low. If a scavenging liquid is used, it must be compatible with the hydraulic fluid. If, however, the hydraulic fluid is immediately used for scavenging, it is an advantage to leave it in the equipment because an oil change may enable fresh impurities to enter the tank. After scavenging new filter elements are to be inserted.

During the first start-up the liquid level falls because the oil distributes itself all over the pipe system of the equipment. Therefore hydraulic fluid must be filled up as quickly as possible. Later on the liquid level must also be checked at regular intervals. The fluid must on no account remain below the lower marking.

3. START-UP PROCEDURES

3.1 PREPARATIONS

- Check oil level in tank.
- Any shut-off valves in the suction pipes must be fully opened.
- Prefill the housings of all piston pumps with clean hydraulic medium.
- Care shall be taken that the equipment can start up operationally reliable and that no operators are within hazard areas.

3.2 START-UP OF THE EQUIPMENT

- All time data, temperature values, and all other details worth knowing should be recorded for the future maintenance of the hydraulic equipment.
- The electric motor is briefly switched on and immediately switched off again. The sense of rotation of the pumps is checked.
- If automatic deaeration is not available, the screwed joint at the pressure side of the pump is loosened or deaeration valves opened so that the pump can start up more easily and air can leave the equipment better.
- The mobile parts of the equipment are to be run under low pressure so that air escapes. This shall be continued until the liquid emerging from the deaeration connections is completely free of bubbles.
- The equipment is to work against low pressure for some time. This is, so to speak, the last scavenging before the pump is started up under full load, always provided that oil flows through the filters of the equipment.
- Pressure valves are to be set to the lowest value the pump can satisfactorily work with.
- If the equipment is equipped with a compensator pump, an additional pressure control valve, then the value set at the pressure control valves must at least be 10 bar higher than the pump pressure.
- Once again the oil level in the pump shall be checked.
- As soon as operating temperature has reached its normal value all screwed pipe connections and mountings must be checked and retightened if necessary.

- Filters shall be cleaned frequently during the first running period and thereafter at intervals taught by experience. Fine filters with a high storage capacity are recommended for the first running period of hydraulic equipments that have to work with high precision.

3.3 HYDRAULIC FLUIDS

3.3.1 Hydraulic oil

The hydraulic media of a hydraulic equipment must have a series of properties in order to be able to guarantee operating safety for a long time. It is recommended to make the choice together with a well-known oil producer. Hydraulic oils contain a lot of additives, eg. with properties inhibiting corrosion and reducing wear. They must comply with DIN 51524. When hydraulic fluids are topped up, one should in principle use the identical fluid the system has been filled with. This should also be considered when using mineral oils, for oils that may comply with DIN 51524, part I or part II, are still differing in their use of additives and their basic oils.

Mixtures of various makes should in any case be avoided with hydraulic fluids of classes HFA, HFC, and HFD.

If fluids of the same type, but of different producers are mixed, any question of liability resulting from damages can no longer be allocated.

It can be said that producers of hydraulic fluids often decline any responsibility if any additives of different producers were mixed into the hydraulic fluid. The manufacturers of hydraulic devices adopt a similar attitude.

Even before using so called HEPG, HETG and HEES "Bio-Oils" manufacturer should be consulted.

In principle only relevantly qualified staff should carry out the start-up of hydraulic equipments. During initial operations the circuit diagram, list of parts, technical information of the installed devices and measuring equipment should be available. Start-up must be carried out with due care. Responsible staff of the manufacturer should be present. The same applies to the personnel of the operator. It is recommended to prepare a protocol of the start-up.

4. UPKEEP AND MAINTENANCE

4.1 MAINTENANCE

Maintenance measures must be planned and carried out in relation to value, action time and the required availability.

4.1.1 Inspection

Various inspection intervals will be required according to different action times eg. daily inspection, monthly inspection, and an inspection directly prior to a longer standstill of the equipment.

Important items are:

- Check liquid level in tank

Normally too low a level of liquid points to loss by outer leakages, after more intensive upkeep work the liquid level may continue to drop slowly for some time if the equipment is automatically deaerated. Too high a liquid level may indicate work without load in the upper parts of the equipment caused by penetration of air during standstill periods.

- Check heat exchanger for effectiveness -

Air - liquid heat exchanger -

Heat loss drops very quickly on account of dust deposits on the surface of the heat exchanger especially in locations with a high development of dust.

If cooling air then contains oil mist from outer leakages as well, heat loss is almost completely interrupted within a very short time.

Water - liquid heat exchanger -

Cooling water must be so clean that the transfer of heat is not hindered by insulating layers of mud. This is of particular importance if the temperature of the hydraulic fluid is kept constant by throttling the cooling water, thus not allowing for high water speeds to flush out deposits. It is important that water heat exchangers can drain neither at the side of the water nor the hydraulic fluid, because only partial surfaces will then be used for heat transmission.

- Check imperviousness of the equipment towards the outside

Pipelines, hosepipes (especially at the place of connection), pumps, control devices, hydraulic motors, and cylinders are to be checked.

- Check liquid temperature during operation

A rise in liquid temperature may have the following reasons:

The heat exchanger has too low an effect (surface soiled, ventilator out of action, cooling water supply insufficient or interrupted, cooling water temperature too high at entry, heat exchanger silted up).

Heat dissipation of the liquid tank, the pipeline system, and the hydraulic components has been reduced by dirt deposits.

Inner leakage of individual components has increased, pressure control valves respond to pressure that is too low. The equipment is operated outside the authorized data, etc.

- **Check pressures**

Check cutting-in pressures of pressure relief valves as well as set pressures of pressure reducing, unloading valves etc..

- **Check leakage quantities**

By measuring the quantities of leakage oil at hydraulic motors and partly at pumps, conclusions can be reached concerning the state of wear. Slow withdrawal (or extension) of a cylinder under outer load with all shut-off devices blocked may indicate defective piston packings.

- **Check noise development**

A hissing noise is heard when pressure valves open. Rattling or whistling noises indicate a defect.

Cylinders may create rattling or grating noises when extended or withdrawn. It may indicate worn guides, secondary bending (eg. rusted-in tilting bearings), unsuitable hydraulic fluid etc.

An increase in pump noises (applies to oil engines, too) under mounting pressure may indicate erosion or cavitation damages within the control surfaces or increasing play of the displacement elements or beginning roller bearing damages. Unpleasantly loud noise of the pumps, independent of the pressure load, point out too high negative pressure in the suction (or supply) pipe.

- **Filter check**

Whenever necessary, filter cartridges must be replaced in reflux filters equipped with contamination indicators. Dirty filters may lead to malfunctions of the hydraulics within a short time.

The same applies to pressure filters. In principle, filter elements are to be cleaned or replaced whenever the liquid is changed.

- **Performance and speed**

Check extension and withdrawal times of cylinders or the speed of oil engines, respectively.

Measure power consumption of electric motors of the pumps.

- **Pipeline system and hosepipes**

Sealing work at the pipeline system only to be carried out in pressureless state. Leakages in places sealed with soft sealings (O-rings, form ring seals) cannot be removed by re-tightening as these sealing elements are either destroyed or hardened.

Sealing can only be carried out by replacing the elements. Pipelines to be checked for firm seat at fixing points.

Hosepipes to be checked for chafe marks and hosepipe fittings for impermeability.

4.1.2 Cleaning the equipment

Hydraulic equipments are to be cleaned on the outside from time to time in order to find leakages, avoid taking dirt into the interior when refilling hydraulic fluid and replacing filter elements, protect cylinder rods against grooving during extension and withdrawal, not to reduce or completely interrupt head dissipation considered in the design.

When cleaning, it always has to be taken care that cleaning fluid does not enter the hydraulic system.

Exchange of hydraulic fluid should be carried out at regular intervals - but at least once a year. At the same time the medium contained in the pipelines and cylinders should also be removed. The hydraulic system is to be deaerated after the renewed start-up.

4.1.3 Maintenance of pressure vessels

The gas precharge pressure of accumulators is to be checked at intervals, always under almost identical ambient temperatures and with a pressureless hydraulic system. This applies to diaphragm accumulators, piston accumulators and bladder accumulators.

For further information see our information sheet 15-007.

With very few exceptions hydraulic accumulators are filled with nitrogen as far as gas is concerned.

For safety reasons a sign with the following, easily legible text must be so applied to all equipments that it cannot be overlooked.

**ATTENTION - PRESSURE
ACCUMULATOR EQUIPMENT!
Prior to starting repair work the
equipment must be rendered pressureless
at the side of the liquid!**

4.1.4 Routine maintenance

The following list contains proposals for some checks to be made at regular intervals:

1. Machine cycle time.
2. Operating temperature of the hydraulic fluid.
3. Reading the individual manometers.
4. Unusual noises.
5. Regular measurements of leakages at the piston housing under identical pressure and temperature conditions to establish comparative values. Changes in these leakages will yield information about the general state of the equipment.

Essential deviations from the nominal values with regard to the above 5 items may lead to further investigations of the causes of these faults and to the implementation of maintenance or upkeep measures.

6. Clean all hydraulic devices on the outside and carry out leakage checks. If necessary, screwed connections must be retightened. If this is no longer feasible, they must be exchanged.
7. Check whether heat exchangers are clean and impervious.
8. Remove hydraulic fluid to check its chemical composition and degree of contamination.
9. Clean filter elements.
If necessary, replace with new ones.
10. Check ventilation filter for dirt deposits. Clean or replace, if necessary.
11. Check whether pre-charge pressure has the proper value when using gas-filled accumulators. Regulations of the manufacturer are to be precisely followed when refilling.

4.2 OVERHAUL OF THE EQUIPMENT

If routine maintenance and inspections show after some years of operation that the performance of the equipment no longer corresponds with the specified standard values, it should be time for a thorough overhaul of the hydraulic devices. This offers the opportunity to really thoroughly clean all the devices, namely the inside of the tank.

The use of a plain scraper of plastic is recommended for cleaning even surfaces, because fibres may be left behind when using cloths. These may lead to considerable breakdowns. It is advisable to get into contact with a firm of experts for hydraulics before starting overhaul work and obtain their advice for maintenance and the replacement of components suspected of faults or wear.

In this context a set of new sealings can also be ordered. This is recommendable as after several years of operation the sealing elements will no longer guarantee proper sealing on account of hardening.

Hoses in the supply lines and tubular feedthroughs that are worn or have become porous should also be checked.

The relevant provisions must be observed when reassembling the hydraulic devices. It has always to be taken care that the hydraulic devices are absolutely clean inside.

4.3 SAFETY MEASURES

Proper care must always be exercised when starting-up or maintaining the equipment. Let us remind you in particular of the following points:

1. Accumulators to be rendered absolutely pressureless when their pipelines are removed.
2. Check whether clutches slip or are loose.
3. Loads that may drop without control or may be moved in any other way during the disassembly of the hydraulic drive or work on the pipeline must be supported or wedged.
4. Before starting any work whatsoever on the equipment it must be guaranteed that the electrical current is switched off.

The best solution is to remove the main fuses.

5. GENERAL HAZARD INFORMATION

5.1. GENERAL REMARKS

Integral Hydraulic manufactures hydraulic components and complete hydraulic systems according to the acknowledged technical rules. They can be considered safe when used pursuant to provisions and properly maintained. On the other hand, hydraulic components and systems provide hazards that may lead to property damages and personal injuries if proper care is not taken.

5.2 HAZARDS

5.2.1 Caused by the pressure medium

Even without pressure the pressure medium may already cause damages by its chemical properties and in addition by the energy immanent under pressure.

5.2.1.1 Fire hazard

Mineral oil generally used is flammable in its liquid form and especially as an oil-air-mist. Therefore oil leakages, handling open flames as well as the presence of parts running hot (also during transport) are to be avoided. If need be, danger warnings are to be affixed.

5.2.1.2 Toxicity

Although hydraulic media are not expressly classified as toxic, any inhalation and prolonged contact with the skin should be avoided. Contaminated clothing should be changed as soon as possible. Affected skin is to be cleaned.

5.2.1.3 Temperature

At high temperatures emerging pressure medium may lead to scalding. If too high a temperature development is suspected, cooling should be provided.

5.2.1.4 Hazards for the environment

Freely leaking pressure medium may contaminate the soil, groundwater and other waters. Therefore free leakage should be avoided by collecting basins etc. Especially in cases of mobile deployment the use of pressure media environmentally compatible ("bio-oils") should be taken into consideration.

5.2.1.5 Hazards by kinetic energy

Under strong differences in pressure escaping jets of liquid may become extremely dangerous by their drilling (injecting) or separating effect. Prevention means the proper dimensioning of all devices, line connections, screwed connections, etc., proper tightening of all screwed connections, safe fixings for all devices, pre-scribed welded connection as well as reliable pressure control taking dynamic influences into consideration as well.

5.2.1.6 Oxygen displacement

This hazard may only occur when larger hydraulic accumulators filled with nitrogen are emptied at the side of the gas in narrow, unventilated rooms. If it should prove to be unavoidable, ventilation must be provided.

5.2.1.7 Consequential damages by lack of compatibility

Lacking compatibility towards sealing materials, bubbles, membranes, coats of paint, etc. may lead to leakages with the hazards described above or even to complete break-down. Therefore only pressure media recommended by the manufacturers should be used. The manufacturer should be consulted in case of other media or extraordinary operating conditions (eg. low minus temperatures).

5.2.2 Unintentional movement by pressure energy

Unintentional movements may be caused either by working devices in the sense of unexpected work-motions or by parts/devices flying away in case of loosening or bursting.

5.2.2.1 Breakdown of controls

Failing of controls should not lead to unintentional movements. Controls should be safe enough to stop or interrupt, in any case, hazardous movements when a control fails (fail-safe concept). Wherever this cannot be achieved, a different type of protection (eg. shields, covers, cut-offs, etc.) must be provided.

5.2.2.2 By accumulated energy

Hydraulic systems with an accumulator effect (hoses, hydraulic accumulators) store energy that may lead to hazardous movement even after the supply of primary energy has been cut. It is to be examined whether the system cannot automatically be rendered pressureless if primary energy fails. If this is not the case, eg. with accumulator-supported clamping fixtures, a clear hazard warning must be available.

5.2.3 By component failure

Bursting, tearing, detaching, loosening, opening, etc. may make fragments or entire devices fly off. It may be caused by static overloading, but even more so by dynamic vibration stresses. Finally outside forces may lead to damages.

6. ORDINANCES, PROVISIONS, TECHNICAL RULES

A small selection of publications and titles concerning safety:

- Directive of the Council of 14 June 1989 for the harmonization of legal provisions for machines of the member states.
- Pressure Vessel Ordinance (DruckbehV) of 27.02.1980 in its latest version (4.96).
- Technical Rules Pressure Vessels (TRB) in their relevant latest version.
- ISO 4413 Hydraulic Fluid Power - General rules for the application to transmission and control systems.
- DIN EN (Acc. to EN 982) Safety of machinery, Safety requirements for fluid power systems and their components, Hydraulics
- DIN 24 552 Hydraulic accumulators in hydraulic systems (with annex).
- Rules for the Prevention of Accidents - power-driven equipment VGB 5 April 1986.

For normal working conditions we recommend usage of HL and HLP Hydraulic oils (on Mineral oil basis). In special cases also HLPD and HVLP oils with same viscosity may be used. Please contact us if you wish to use HE-Liquids (HEPG, HETG and HEES oils) or HFC (not easily inflammable water – Glycole mixture).

The sequence of the listed suppliers is alphabetical and therefore no standard of value. We would also like to indicate that the list is not complete and that we cannot guarantee for it's context. Please take notice that some liquids cover a wide range of viscosity and therefore are only mentioned once (e.g. 32-68).

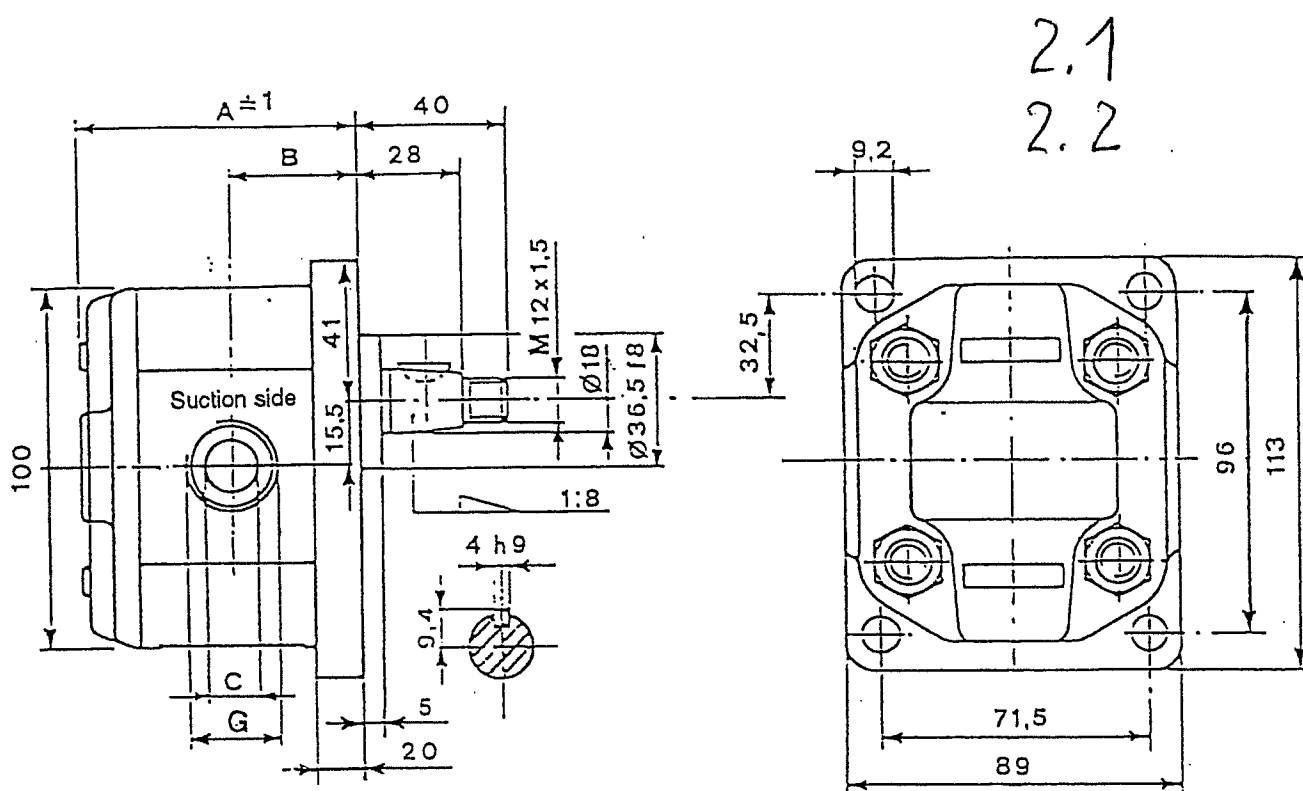
- 1) normally Hydraulic Oil acc. to DIN 51524 part 1
- 2) normally Hydraulic Oil acc. to DIN 51524 part 2
- 3) normally Hydraulic Oil acc. to DIN 51524 part 3
- 4) like ²⁾ or ³⁾ but detergent

Sheet 1 of 2

Oil company (alphabetical order)	VG 46		VG 68	
	a) HL ¹⁾ (mineral oil) b) HLP ²⁾ (mineral oil) c) HVLP ³⁾ (mineral oil) d) HLPD ⁴⁾ (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)	a) HL (mineral oil) b) HPL (mineral oil) c) HVLP (mineral oil) d) HLPD (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)
ARAL	a) Aral Vitam UF 46 b) Aral Vitam GF 46 c) Aral Vitam HF 46 Aral Vitam VF 46 d) Aral Vitam DE 46	e) Aral Vitam BAF 45 g) Aral Vitam EHF 45 h) Aral Montral 44	a) Aral Vitam UF 68 b) Aral Vitam GF 68 c) Aral Vitam VF 68 d) Aral Vitam DE 68	
BECHEM	b) Staroil Nr. 46 c) Staroil HVI 46 d) Staroil H-LPD 46	e) Hydrostar UWF 45 f) UWS Hydraulik 32 g) Hydrostar HEP 45 h) Hydrostar HY 46	b) Staroil Nr. 68 c) Staroil HVI 68 d) Staroil H-LPD 68	e) Hydrostar UWF 68 g) Hydrostar HEP 68
BP	a) BP Energol HL 46 b) BP Energol HLP-HM 46 c) Bartran HV 46 d) BP Energol HLP-D 46	f) Biohyd 46 g) Biohyd SE 46 h) Enersyn SF-C 14	b) BP Energol HLP-HM 68 c) Bartran HV 68 d) BP Energol HLP-D 68	g) Biohyd SE 68
CASTROL	a) Magna 46 b) Hyspin AWS 46 Hyspin SP 46 c) Hyspin AWH-M 46 d) Vario HDX 46 Hydrauliköl HLP-D 46 SF	f) Carelube HTG 32 g) Castrol Produkt 695/13 Carelube HES 46 h) Anvol WG 46	a) Magna 68 b) Hyspin AWS 68 Hyspin SP 68 c) Hyspin AWH-M 68 d) Vario HDX 46 Hydrauliköl HLP-D 68 F	f) Carelube HTG 68 g) Castrol Produkt 695/14 Carelube HES 68
DEA	a) Astron HL 46 b) Astron HLP 46 Astron X HLP 46 c) Astron HVLP 46 Astron X HVLP 46 d) Actis HLPD 46 Actis X HLPD 46 Trion EP 46	e) Econa PG 46 f) (Econa R 32) g) Econa E 46 h) Tectro HF-C 46 S	a) Astron HL 68 b) Astron HLP 68 Astron X HLP 68 c) Astron HVLP 68 d) Actis HLPD 68 Actis X HLPD 68 Trion EP 68	
ELF	a) ELF POLYTELIS 68 b) ELFOLNA 68 ELFOLNA DS 68 ELFOLNA SP 68 c) HYDRELF DS 68 d) ELFOLNA HLPD 68 ELFOLNA HMD 68	f) ELF XTD 93031 g) HXDRELF BIO h) PYRELF HFC 46	a) ELF POLYTELIS 46 b) ELFOLNA 46 ELFOLNA DS 46 ELFOLNA SP 46 c) HYDRELF DS 46 d) ELFOLNA HLPD 46 ELFOLNA HMD 46	
ESSO	a) TERESSO 46 b) NUTO H 46 Hydrauliköl HLP 46 c) UNIVIS N 46 d) HLPD-OEL 46	e) Hydrauliköl PGK 45 f) Hydrauliköl PFL g) Hydrauliköl HE 46	a) TERESSO 68 b) NUTO H 68 Hydrauliköl HLP 68 c) UNIVIS N 68 d) HLPD-OEL 68	
FINA	a) CIRKAN 46 b) HYDRAN 46 c) HYDRAN HV 46 d) HYDRAN HLP-D 46 Hydrauliköl D3033	e) Hydrauliköl D3031-46 f) BIOHYDRAN RS 38 g) BIOHYDRAN SE 38 BIOHYDRAN TMP 46	a) CIRKAN 68 b) HYDRAN 68 c) HYDRAN HV 68 d) HYDRAN HLP-D 68	g) BIOHYDRAN TMP 68

Sheet 2 of 2

	VG 46		VG 68	
Oil company (alphabetical order)	a) HL ¹⁾ (mineral oil) b) HLP ²⁾ (mineral oil) c) HVLP ³⁾ (mineral oil) d) HLPD ⁴⁾ (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)	a) HL (mineral oil) b) HLP (mineral oil) c) HVLP (mineral oil) d) HLPD (mineral oil)	e) HEPG (polyglykol-basis) f) HETG (vegetable oil) g) HEES (synth. ester) h) HFC (water/glycol)
FUCHS	a) RENOLIN DTA 46 b) RENOLIN B15 VG 46 RENOLIN ZAF 46 B c) RENOLIN MR 46 MC RENOLIN ZAF 46 MC d) RENOLIN MR 15 VG 46 RENOLIN D 15 VG 46 RENOLIN ZAF 46 D	e) RENOLIN PG 46 f) PLANTOHYD 46 N g) PLANTOHYD 46 S PLANTOHYD Super S h) Hydrotherm 46 M Hydrotherm 46 NF3	a) RENOLIN DTA 68 b) RENOLIN B20 VG 68 RENOLIN ZAF 68 B c) RENOLIN MR 68 MC RENOLIN ZAF 68 MC d) RENOLIN MR 20 VG 68 RENOLIN ZAF 68 D	f) PLANTOHYD 68 N g) PLANTOHYD 68 S
MOBIL	a) Vactra Oil Medium DTE Oil Medium b) Mobil DTE 25 c) Mobil DTE 15 M d) Hydrauliköl HLPD 46	f) Mobil EAL 224 H g) Mobil EAL Syndraulic 46 Hydraulic Oil UF 46 h) Hydrofluid LT Nyvac FR 200 D Fluid	a) DTE Oil Heavy Medium Vactra Oil Heavy Medium b) Mobil DTE 26 c) Mobil DTE 16 M d) Hydrauliköl HLPD 68	
OEST	a) Hydrauliköl H-L 46 b) Hydrauliköl H-LP 46 c) Hydrauliköl HVI 46 d) Hydrauliköl 46 DD	f) (BIO HY-FLUID HV 34) g) BIO Synthetik HYD 45	a) Hydrauliköl H-L 68 b) Hydrauliköl H-LP 68 c) Hydrauliköl HVI 68 d) Hydrauliköl 68 DD	f) BIO HY-FLUID HV 68 g) BIO Synthetik HYD 68
PANOLIN	a) Panolin Indol ISO 45 b) Panolin HLP ISO 46 c) Pan. HLP Universal 37 d) Panolin HLP-D ISO 46	g) Panolin HLP Synth 45	a) Panolin Indol ISO 68 b) Panolin HLP ISO 68 c) Panolin GP 55 d) Panolin HLP-D ISO 68	g) Panolin HLP Synth 68
PETROFER	a) Isolubric VG 46 L b) Isolubric VG 46 c) Isolubric VG 46 HV d) Isolubric VG 46 D	e) Syntolubric 46 f) Tribolubric 32 g) Envolubric HE 46 h) Ultra-Safe 620	a) Isolubric VG 68 L b) Isolubric VG 68 c) Isolubric VG 68 HV d) Isolubric VG 68 D	g) Envolubric HE 68 h) Ultra-Safe 360
QUAKER		f) GREENSAVE N 30 g) GREENSAVE N 40 h) QUINTOLUBRIC 730		
SHELL	a) Morlina Oil 46 b) Shell Tellus Oil 46 c) Shell Tellus Oil TD 46 d) Shell Tellus Oil DO 46	e) Shell Fluid BD 46 f) Shell Naturelle HF-R g) Shell Naturelle HF-E 45 h)	b) Shell Tellus Oil 68 d) Shell Tellus Oil DO 68	g) Shell Naturelle HF-E 68
STUART		e) ISOCOR E 46 g) ISOCOR HF 46 h) HYDROCOR CC 44		g) ISOCOR HF 68
TEBIOL		f) Florahyd HVI 46 g) Esterhyd HE 46		f) Florahyd HVI 68
TRIBOL	a) Tribol 772 b) Tribol 943 AW 46		a) Tribol 773 b) Tribol 943 AW 68	
WISURA	a) Dynex 46 b) Tempo 46 c) Hydroma 46 d) HLPD 46	f) Hydroma NAT 40 g) Hydrofluid SE 46	a) Dynex 68 b) Tempo 68 c) Hydroma 68 d) HLPD 68	



Direction of rotation:

Installation position:

Pipe connections for inlet and outlet port::

System pressure range, input p_1 :

System pressure range, output p_2 :

Output flow:

Temperature range :

Hydraulic oil:

Viscosity at + 50°C:

Viscosity min./max.:

Filtration::

Shaft load:

Clockwise (counterclockwise on request)

Optional

Thread

$p_{1 \min} = 0,7 \text{ bar (10 PSI) absolute}$

$p_{1 \max} = 2,5 \text{ bar (35 PSI) absolute}$

Maximum pressure = see chart

see back side

min. -10°C max. +85°C

see data sheet 01-12.E

Recommended 25 to 50 mm²/s

20 mm²/s / 1000 mm²/s (intermittent cold start)

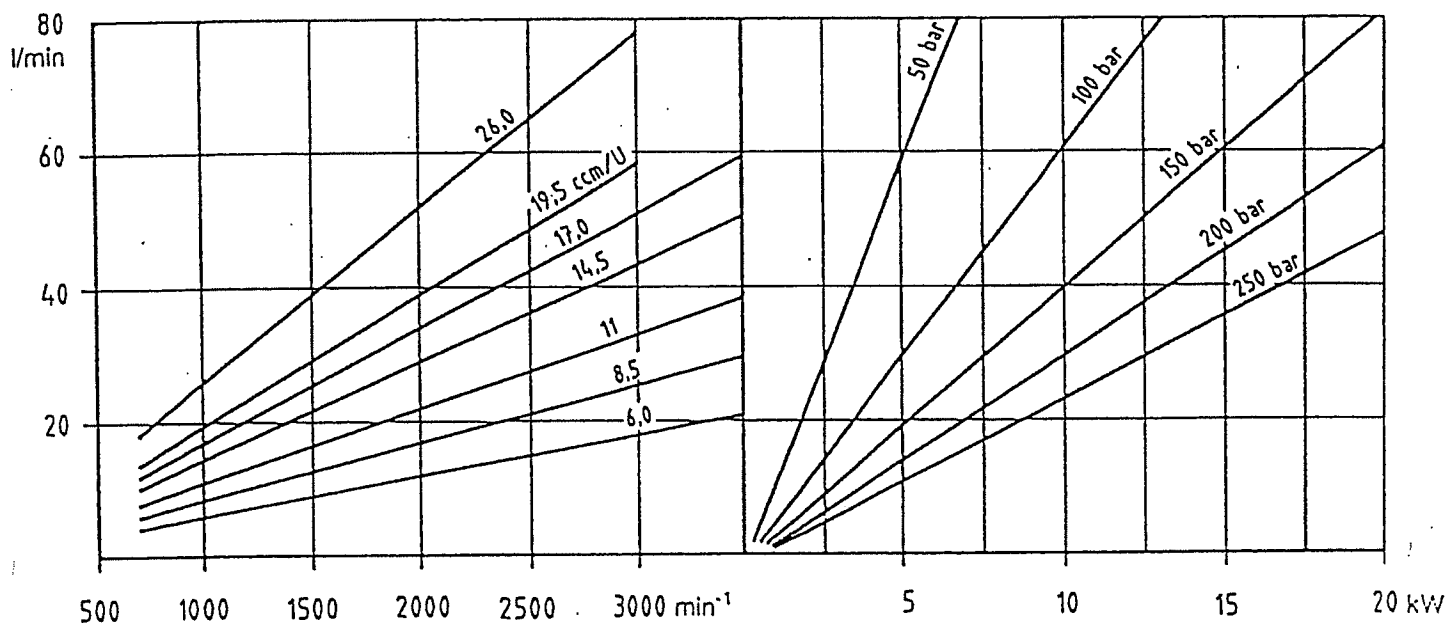
Return filter 15 to 20 µm

The drive must not impose severe axial or radial loads on the pump shaft. An oldham coupling or similar three-piece drive arrangement is recommended. Indirect drives require independent front bearing supports, which are available on request.

Type	Order No.	Output flow at n = 1000 rpm		Max. pressure [bar]	Speed [rpm]		A [mm]	B [mm]	Screw in thread				Weight	
		[l/min]	[gpm]		max.	min.			Suction connection	Pressure connection	[kg]	[lb]		
MLP D/G 206 C	060-1323-016-270	6	1,6	270	3 500	700	98	46	G	C	G	C	2,4	5,28
MLP D/G 208 C	085-1323-016-270	8,5	2,3	270					G 1/2	13,5	G 1/2	13,5		
MLP D/G 211 C	110-1323-016-270	11	2,9	270										
MLP D/G 214 C	145-1323-016-260	14,5	3,9	260			120	58	G 3/4	20	G 1/2	13,5	2,8	6,16
MLP D/G 217 C	170-1323-016-250	17	4,5	250										
MLP D/G 220 C	195-1323-016-220	19,5	5,2	220			3 000	133	64,3	G 1	20	G 3/4	20	3,1
MLP D/G 226 C	260-1323-016-190	26	6,9	190	3 000									

Technical Information:

Theoretical flow - Pump revolutions
Flow - Theoretical power



Total efficiency:

measured bei working pressure 200 bar and 1 500 rpm
($\nu = 35 \text{ mm}^2/\text{s}$; $\vartheta = 50^\circ\text{C}$)

Pump [l/min]	η_v	$\eta_{\text{ges.}}$
6 ... 26	0,96	0,88 ... 0,92

Power (calculating relations)

$P = Q \cdot \Delta p / 612 \cdot \eta_{\text{ges.}}$
 P = absorbed power in kW
 Q = capacity in l/min
 Δp = working pressure in bar
 $\eta_{\text{ges.}}$ = total efficiency for pump (see tables)

Limits:

In order to obtain Optimum performance, the number of revolutions should not exceed the maximum values shown in the tables. The pressure values $P_{\text{max.}}$ shown in the tables are for continuous running at 1 500 rpm revolutions with nominal hydraulic oil having a viscosity of 25 ... 50 mm^2/s . For different viscosity (high temperature) or revolution values it is necessary to decrease continuous operating pressure.

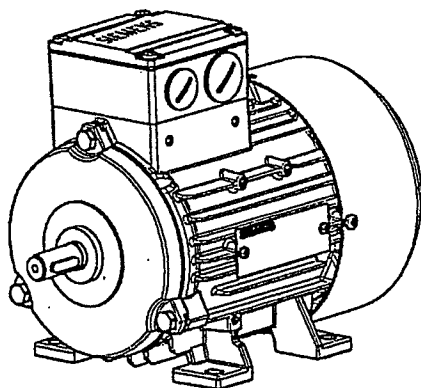
Oil viscosity 12 mm^2/s for 1 500 rpm max. ED 20s: 85% von $P_{\text{max.}}$
 or ED 2s: 90% von $P_{\text{max.}}$

Oil viscosity 20 ... 50 mm^2/s for 1 500 rpm ED 100%: $P_{\text{max.}}$
 $< 1\,500$ and $> 2\,000$ rpm ED > 20 s: 85% $P_{\text{max.}}$

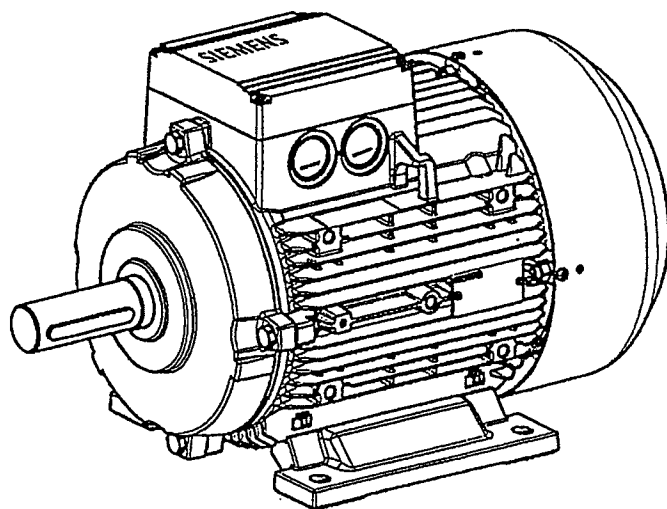
Drehstrommotoren Three-phase AC Motors Moteurs triphasés Motores trifásicos Motori trifasi Trefasmotorer

Betriebsanleitung
Instructions

1LA7^{CE}, 1LP7^{CE}, 1PP7^{CE}
1MA7, 1MF7



BG 71 ... 90 L - IM B3



BG 100 ... 160 L - IM B3

Baugrößen (BG) / Frame sizes (BG) / Désignation de carcasse (BG)
Tamaños constructivos (BG) / Grandezza costruttiva (BG) / Konstruktionsstorlekar (BG)

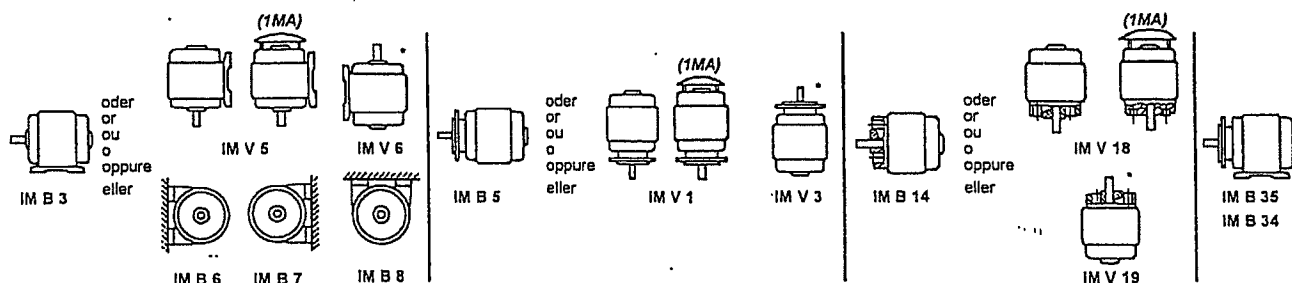


Fig. 1 Bauformen / Types of construction / Formes de construction
Formas constructivas / Forme costruttive / Monteringsätt

* s. "4 EEx e-Motoren" / see "4 EEx e-motors" / voir "4 Moteurs EEx e" / véase "4 Motores EEx e" / cfr. "4 Motori EEx e" / se "EEx e-Motorer"

gegen unzulässige Erwärmung zu schützen. Die Schutzeinrichtung ist auf den Bemessungsstrom einzustellen. Bei Wicklungen in Dreieckschaltung werden die Auslöser in Reihe mit den Wicklungssträngen geschaltet und auf den 0,58fachen Bemessungsstrom eingestellt. Ist diese Schaltung nicht möglich, sind zusätzliche Schutzmaßnahmen erforderlich (z. B. Thermischer Maschinenschutz).

Die Schutzeinrichtung muß bei blockiertem Läufer innerhalb der für die jeweilige Temperaturklasse angegebenen t_E -Zeit abschalten.

Elektrische Maschinen für Schweranlauf (Hochlaufzeit $> 1,7 \times t_E$ -Zeit) sind entsprechend den Angaben der Konformitätsbescheinigung durch eine Anlaufüberwachung zu schützen.

Thermischer Maschinenschutz durch direkte Temperaturüberwachung der Wicklung ist zulässig, wenn dies bescheinigt und auf dem Leistungsschild angegeben ist.

Bei polumschaltbaren Motoren sind für jede Drehzahlstufe getrennte, gegenseitig verriegelte Schutzeinrichtungen erforderlich. Empfohlen werden Einrichtungen mit dem Prüfbericht einer zugelassenen Prüfstelle.

In Deutschland wird beim Errichten elektrischer Anlagen in explosionsgefährdeten Bereichen auf DIN 57165/VDE 0165 und ExeV hingewiesen! Im Ausland sind die entsprechenden Landesvorschriften zu beachten!

Der Betrieb am Umrichter muß ausdrücklich bescheinigt sein. Die gesonderten Herstellerhinweise sind unbedingt zu beachten. Für die Zündschutzart EExe müssen Motor, Umrichter und Schutzeinrichtungen als zusammengehörig gekennzeichnet und die zulässigen Betriebsdaten in der gemeinsamen Prüfbescheinigung festgelegt sein (VDE 0165).

Die vom Umrichter erzeugten Spannungsspitzen können durch das installierte Verbindungskabel zwischen Umrichter und elektrischer Maschine in ihrer Größe ungünstig beeinflusst werden. In dem System Umrichter-Kabel-elektrische Maschine darf der Maximalwert der Spannungsspitzen an den Anschlußklemmen der Maschine den in den gesonderten Herstellerhinweisen genannten Wert nicht überschreiten. Des weiteren ist die EMV-Richtlinie einzuhalten.

Reparaturen müssen in Siemens-Werkstätten durchgeführt oder von einem amtlich anerkannten Sachverständigen abgenommen werden. Die Arbeiten sind durch ein zusätzliches Reparaturschild zu kennzeichnen.

Ersatzteile mit Ausnahme genormter, handelsüblicher und gleichwertiger Teile (z. B. Wälzlager) dürfen nur Originalersatzteile (s. Ersatzteilliste) verwendet werden; dies gilt insbesondere auch für Dichtungen und Anschlußteile.

General note



The data and recommendations specified in all the instructions supplied ("Information on safety and commissioning"), and in all other related instructions, must always be observed in order to avoid hazardous situations and the risk of possible injury or damage.

Furthermore, the pertinent national, local and plant-specific regulations and requirements should be kept in mind!

Special designs and other versions may vary in technical details! If in doubt, be sure to contact the manufacturer, quoting the type designation and serial number (No. E ..., see rating plate), or have maintenance work done by one of the SIEMENS Service Centres.

1 Description

1.1 Application

Normal use of the standard motors:

The motors are designed to comply with degree of protection IP55 (degree of protection indicated on rating plate). They are suitable for installation in dusty and damp environments. The insulation is tropicalized. If they are properly stored or installed outdoors, special weatherproofing measures are not usually required for these motors. They must however be protected against intense sunlight, e.g. by means of a canopy.

Ambient temperature $-30^{\circ}\text{C} \dots +60^{\circ}\text{C}$

Site altitude $\leq 1000 \text{ m}$

1.2 Construction and mode of operation

Motor types:

1LA- and 1MA motors are self-ventilated (with fans). 1PP motors are self-ventilated either by separate fans arranged on the rotors, such as for service as fan motors, or by external fans.

1LP and 1MF motors are non-ventilated (without fans).

The feet on foot-mounted motors are bolted onto the frame (see Fig. 2).

For this reason, rearranging the feet (such as for changing the position of the terminal box - see Fig. 5) is only possible with frame sizes 100L to 160L. In this case, threads will have to be cut in the existing drilled holes in the frame to receive the feet. However, the support faces of the feet may have to be remachined or provided with shims to ensure that the motor is level.

Where brake motors are concerned, please also observe the brake operating instructions.

2 Operation



Before starting any work on the machine, be sure to isolate it from the power supply.

2.1 Transport, storage

The motors should always be lifted at all lifting eyes during transport.



For lifting machine sets (such as built-on gearboxes, fan units), always use the lifting eyes or lifting pegs provided! Machine sets may not be lifted by suspending the individual machines! Check the lifting capacity of the hoist!

If, after delivery, the motors are stored for more than 4 years under favourable conditions (kept in a dry place free from dust and vibration) prior to commissioning, the rolling-contact bearings should be regreased. Under unfavourable conditions, this period is considerably shorter.

If necessary, any unprotected, machined surfaces (flanging surface, shaft end, etc.) should be treated with an anti-corrosion agent.

If necessary, the insulation resistance of the winding should be checked, see Section 2.5.

2.2 Installation

After installation, screwed-in lifting eyes should either be removed or tightened down.

In the case of motors installed vertically, additional measures must be taken to ensure that no water can penetrate along the shaft.

Quiet running: Exact alignment of the coupling and a well-balanced transmission element (coupling, pulleys, fans, etc.) are essential for quiet vibration-free running. If necessary, the whole motor and transmission element should be balanced.

BG90S/L: The rear feet of motors of type of construction IM B3 have double holes in order to ensure that these machines adhere to the standard foot dimensions. On the other hand, all feet of motors of type of construction IM B3 with side-mounted terminal box (special designs) have double holes. This enables the standard hole clearances to be maintained even when the terminal box is arranged on the left-hand side (see Fig. 10).

BG71...90L: In the case of 1LA, 1LP and 1LP motors, the top part of the terminal box can be turned through 4x90 degrees.

BG100...160L: The terminal box moulded onto the motor frame cannot be turned. If the top part of the terminal box is retrofitted (Fig. 2 - 5.90), the box can be turned through 4 x 90 degrees.

In the case of motors with a bolted-on terminal box, the top part of the box can be turned through 4 x 90 degrees.

2.3 Balancing, transmission elements

A suitable device should always be used for fitting and removing the transmission elements (couplings, pulleys, pinions, etc.) (Fig. 7).

As standard, the rotors are dynamically balanced with the half featherkey.

When fitting the transmission element, keep the type of balance in mind! (Option: F = balanced with full featherkey)

The transmission elements must be balanced in accordance with ISO 1940!

If the transmission elements are balanced with a half featherkey, the visible, protruding part of the featherkey T_p must be cut back (see Fig. 9).



The usual measures should be taken to guard transmission elements from touch. If a motor is started up without the transmission element attached, the featherkey should be secured to prevent it being thrown out.

2.4 Electrical connection

NOTE: If the openings for cables and leads in the terminal box are sealed with a "skin" of cast iron, it must be shaken out using a suitable tool (see Fig. 4)!

Care must be taken not to damage the terminal box, the terminal board, the cable connections, etc. inside the terminal box!

See Fig. 4 for details of the screwed connections for cables and leads. An adapter must be screwed in for PTC thermistor connections!

The terminal box must be sealed so that it is dust and water-tight.

The system voltage and the frequency must agree with the data given on the rating plate. Voltage or frequency deviations of $\pm 5\%$ and $\pm 2\%$ respectively from the rated voltage and frequency values are permitted without needing to derate the output. The connection and arrangement of the terminal links must agree with the diagram provided in the terminal box.

Connect the earthing conductor to the terminal with the  marking.

Wherever terminal clips are used (for example, to DIN 46282), arrange the conductors so that the clips are virtually level on both sides. This method of connection means that the ends of single conductors must be bent in the shape of a U or be fitted with a cable lug (see Fig. 6.1). This also applies to the green-yellow protective earthing conductor and the outer earthing conductor (see Fig. 6.2).

Please refer to Fig. 5 for the tightening torques for the screwed electrical connections - terminal board connections (except for terminal strips).

The anti-condensation heater must not be switched on during operation.

2.5 Checking the insulation resistance

The insulation resistance of the windings must be measured prior to initial startup of the motor, or after long periods of storage or standstill (approx. 6 months).

While the measurement is being taken and immediately afterwards, some of the terminals carry dangerous voltages and must not be touched.

Insulation resistance

- The minimum insulation resistance of new, cleaned or repaired windings with respect to ground is 10 Mohms.
- The critical insulation resistance R_{crit} is calculated first by multiplying the rated voltage U_N , e.g. 0.69 kV AC, with the constant factor (0.5 Mohms/kV):

$$R_{crit} = 0.69 \text{ kV} \cdot 0.5 \text{ Mohms/kV} = 0.345 \text{ Mohms.}^2$$

Measurement

The minimum insulation resistance of the windings to ground is measured with 500 V DC. The winding temperature should then be $25^\circ\text{C} \pm 15^\circ\text{C}$.

The critical insulation resistance should be measured with 500 V DC with the winding at operating temperature.

Checking

If the minimum insulation resistance of a new, cleaned or repaired motor, which has been stored or at standstill for a prolonged period of time, is less than 10 Mohms, this may be due to humidity. The windings must then be dried.

After long periods of operation, the minimum insulation resistance may drop to the critical insulation resistance. As long as the measured value does not fall below the calculated value of the critical insulation resistance, the motor may continue in operation. If it does, the motor must be stopped immediately.

The cause must be determined, and the windings or winding sections repaired, cleaned or dried as necessary.

2.6 Commissioning

NOTE: Electromagnetic compatibility

Emitted interference: Where the torque is very uneven (the drive of a piston-type compressor, for example), the inevitable result is a non-sinusoidal motor current, whose harmonics can lead to excessive system perturbation and thus excessive emitted interference.



In the case of converter-fed motors, interference is emitted to a greater or lesser degree, depending on the converter version concerned (type, interference suppression measures, manufacturer). The instructions of the converter manufacturer regarding electromagnetic compatibility must be heeded at all times. If the use of a shielded motor cable is recommended, the shield will have the greatest effect if it is conductively connected over a large area on the metal terminal box of the motor (with a screwed metal conduit thread). Noise voltages may occur on the sensor leads of motors with integrated sensors (e.g. PTC thermistors) as a result of the converter.

Noise immunity: If the motor has an integrated sensor (e.g. a PTC thermistor), the owner is responsible for ensuring adequate noise immunity by choosing a suitable sensor signal lead (possibly with shielding, connected like the motor supply lead) and evaluator.

The data and recommendations specified in all the instructions supplied ("Information on safety and commissioning"), and in all other related instructions, must always be observed prior to commissioning!

After motor installation, the brake, if fitted, should be checked for proper functioning.

3 Maintenance

Safety precautions



Before starting any work on the motor or other equipment, particularly before opening covers over live parts, the motor must be properly isolated from the power supply. Besides the main circuits, any additional or auxiliary circuits that may be present must also be isolated.

The usual "5 safety rules" (as set forth in DIN VDE 0105) are:

- Isolate the equipment
- Take effective measures to prevent reconnection
- Verify equipment is dead
- Earth and short-circuit
- Cover or fence off adjacent live parts

The precautions listed above should remain in force until all maintenance work is finished and the motor has been fully assembled.

NOTE: Where motors are fitted with closed condensed water openings, these should be opened from time to time to allow any accumulated condensed water to drain away.

Condensed water openings should always be at the lowest point of the motor!

Fitting new bearings, type of grease

Under normal operating conditions, with horizontally mounted motors and the following coolant temperatures and motor speeds, the bearings should be changed at the intervals [h] specified below:

Frame sizes	25°C	40°C
71...160L:	... 1800 rev/min approx. 40,000 h	... 20,000 h
	... 3600 rev/min approx. 20,000 h	... 10,000 h

NOTE: The permissible axial and transverse forces (see Catalog) must not be exceeded!

Irrespective of the number of operating hours, the rolling-contact bearing should be renewed every 3 years because of grease ageing.

In the case of motors operating under special conditions, such as a vertical motor position, heavy vibration, sudden load changes, frequent reversing operation, etc., the bearing should be changed at considerably more frequent intervals than the operating hours stated above.

The motors feature deep-groove ball bearings which are provided with cover plates (2ZC3 version). The cover plate material should withstand temperatures from -30 °C to +150 °C, e.g. polyacryl-rubber (ACM).

Type of grease for standard machines: UNIREX N3 (Esso); synthetic greases must conform to DIN 51825-K3N.

Special greases should be indicated on the rating plate or on a separate plate.

Dismantle the motor to the extent necessary. Pull off the rolling-contact bearing with a suitable device (see Fig. 7). Clean the journal!

Heat the rolling-contact bearing evenly to about 80-100 °C and press on. Heavy blows (such as with a hammer, etc.) should be avoided.

Any worn sealing elements (such as the shaft sealing ring, etc.) should also be renewed.

If springless radial shaft sealing rings are used, the replacement sealing rings must also be of the springless type.

Regreasing device

In the case of motors with a regreasing device, take note of the information given on the rating plate or the lubrication instruction plate! The bearings should be relubricated while the motor is running!

4 1MA and 1MF motors with increased EEx protection



The information in italics is intended to serve as supplementary or special information on these types of motors.

The increased hazards in areas which are exposed to the danger of explosion or firedamp necessitate that the general notes on safety and commissioning are carefully complied with.

Electrical machines which are protected against explosion are in line with the standards EN 60034 (VDE 0530) and EN 50014 to 50020. It is permitted to use these machines in areas exposed to the danger of explosion only in accordance with the stipulations of the responsible authority which also determines whether a danger of explosion exists (division into zones).

If the relevant certification is supplemented by an X, any special stipulations in the certificate of conformity are to be complied with.

The cable entries must be approved for the explosion-endangered area and be secured to prevent accidental loosening. Unused openings are to be closed with approved plugs.

In the case of 1MA-motors, a cover to prevent foreign bodies from falling into the motor-fan cowl (see DIN EN 50014) is to be located on the end of the shaft at the top, e.g. types of construction IMV3, IMV6 and IMV19 (see Fig. 1). This cover must not hinder cooling of the motor by its fan.

Unless other information regarding type of operation and tolerances is given on the test certificate or on the rating plate, electrical machines are designed for continuous operation and normal, rarely recurring starting procedures during which no excessive temperature increases occur. The motors must only be used for the type of operation indicated on the rating plate.

Section A in EN 60034-1 (VDE 0530, Part 1) - voltage $\pm 5\%$, frequency $\pm 2\%$, waveform, power-system symmetry - must be complied with in order to ensure that any increase in temperature remains within the permitted limits. Greater deviations from the rated values can

lead to non-permissible increases in the temperature of the machine and must be indicated on the rating plate.

The temperature class of the motor given on the rating plate must agree with the temperature class of the inflammable gas which may occur.

Each machine is to be protected in all phases against non-permissible temperature increases by means of a current-dependent, time-delayed circuit-breaker with phase-failure protection to VDE 0660 or an equivalent device. The protective device is to be set to the rated current. In the case of delta-connected windings, the trips are to be connected in series with the winding phases and set to 0.58 times the rated current. If such a circuit is not possible, additional protective measures are necessary (e.g. thermal machine protection).

If the rotor is blocked, the protection device must switch off the machine within the time t_g indicated for the respective temperature class.

Electrical machines for heavy starting (power-up time $> 1.7 \times t_g$) are to be protected according to the stipulations of the certificate of conformity by means of a starting-cycle monitoring circuit.

Thermal machine protection by means of direct temperature monitoring of the winding is permissible if this is certified and indicated on the rating plate.

In the case of pole-changing motors, separate and reciprocally locked protective devices are necessary for each speed step. Devices with the test report of an approved testing agency are recommended.

In Germany, please refer to DIN 57165/VDE 0165 and ElexV when erecting electrical installations in areas exposed to the danger of explosion. In countries other than Germany, the relevant national regulations are to be complied with in each case!

Operation on a converter must be certified. The separate instructions of the manufacturer must be complied with. For EEx type of protection, the motor, converter and protective devices must be marked to show they belong together and the permissible operating data must be specified in the shared test certificate (VDE 0165).

S10/3

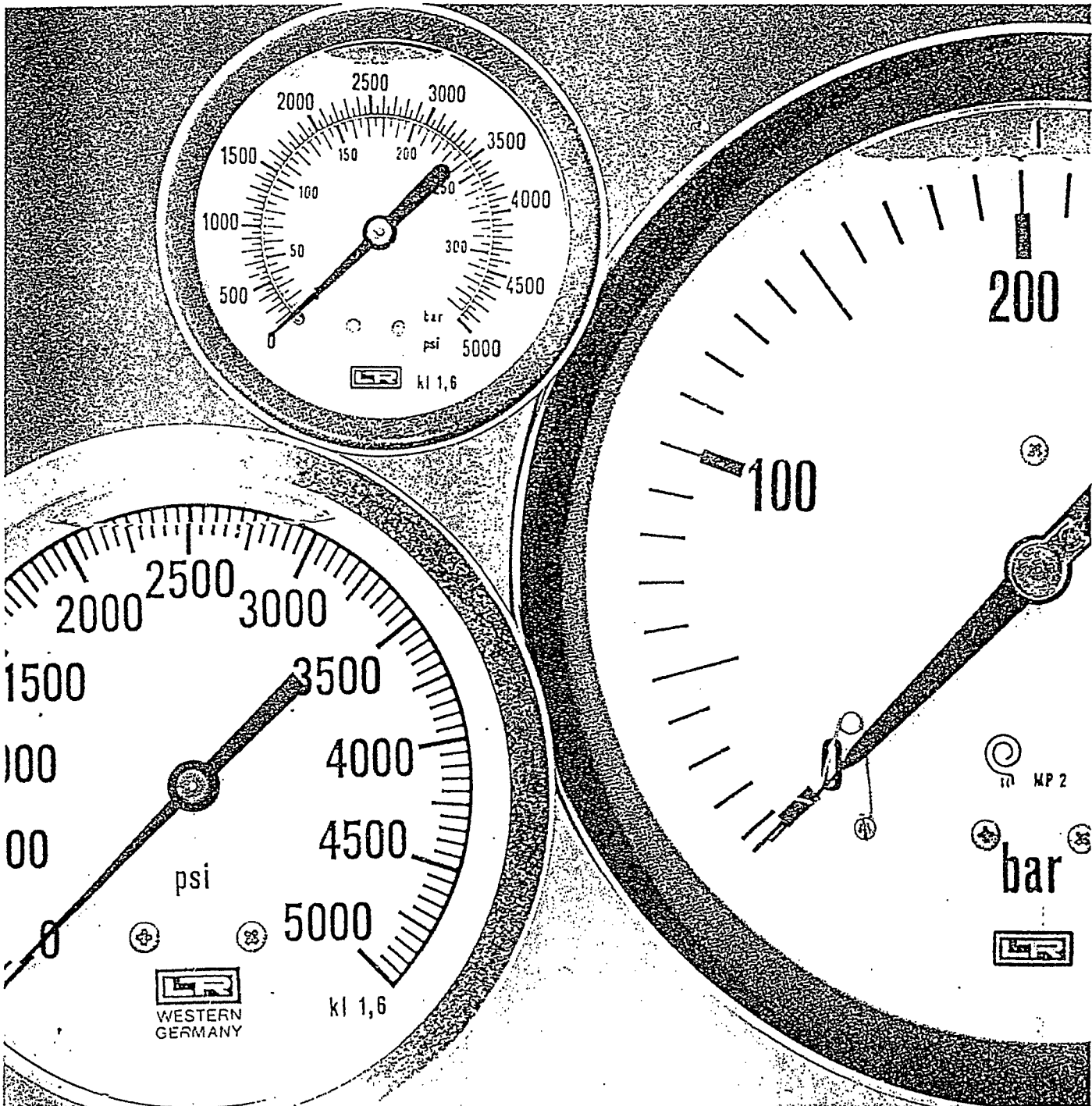
S18/3

1.5

MGS 10/3

4.1.
4.2.

liquid-filled vibration-damping pressure gauges



MATERIAL SPECIFICATION

TYPE S10/3

STANDARD VERSION

Socket material: OT 58 brass.

Elastic element: phosphor bronze or AISI 316L stainless steel.

Movement: gearing, OT 59 brass - plates AISI 304 stainless steel.

Dial: white aluminium, with black markings.

Special Dial: ranges different from standards, or custom artwork, available on application.

Pointer: black oxidized aluminium.

Case: AISI 304 stainless steel.

Gasket: polychloroprene.

Window: plexiglas.

Ring: AISI 304 stainless steel polished, crimped.

Protection: IP 65.

TYPE S18/3

STAINLESS STEEL VERSION TYPE S4

Socket material: AISI 316 stainless steel.

Elastic element: AISI 316L stainless steel.

Movement: reinforced of stainless steel our standard MP2R for DS 100 (4 in.) and DS 150 (6 in.); stainless steel for DS 50-63 (2 - 2 1/2 in.).

Dial: white aluminium, with black markings.

Special Dial: ranges different from standards, or custom artwork, available on application.

Pointer: balanced aluminium or stainless steel; micrometric adjustable only for DS 100 (4 in.) and DS 150 (6 in.).

Case: AISI 304 stainless steel.

Gasket: polychloroprene.

Window: plexiglas; optional: shatterproof glass for DS 100 (4 in.) and 150 (6 in.).

Ring: AISI 304 polished stainless steel; with bayonet connection for DS 100 (4 in.) and DS 150 (6 in.); crimped for DS 50-63 (2 - 2 1/2 in.).

Protection: IP 65.

TYPE S6

Socket material: K Monel.

Elastic element: K Monel.

Window: plexiglas.

Other specifications identical type S4.

DAMPING LIQUIDS

TAB. 1

Damping liquids	Limit ambient temperature	
Glycerine filled 98%	15/65 °C	60/150 °F
Silicone oil filled	-45/65 °C	-50/150 °F
Fluorolube filled	-60/65 °C	-76/150 °F

Glycerine or silicone should not be used with highly oxidizing agents such as oxygen, chlorine, nitric acid or hydrogen peroxide, because of danger of spontaneous

chemical reaction, inflammability or explosion. The use of fluorinated fluid is recommended in these cases.

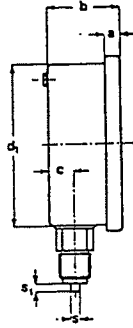
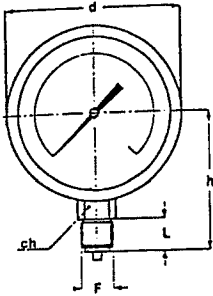
DIAMETERS, ORDERING CODE AND CONNECTING THREADS

TAB. 2

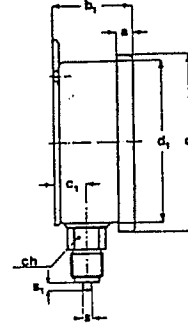
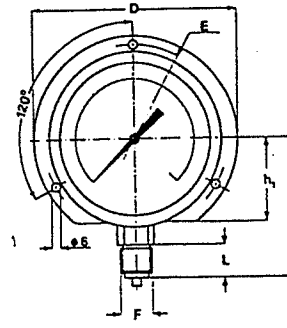
DS	Model	Pressure range (bar)	Ordering Code	Connections	Socket material	C-shaped tubular element	Spiral tubular element	Welding
50	S10/3	2,5/40	N1	1/4"-1/8" Gas-NPT	OT58brass	phosphor bronze	—	Tin alloy
50	S10/3	60/400	S1	1/4"-1/8" Gas-NPT	OT58brass	—	phosphor bronze	Tin/Silver alloy
50	S18/3	2,5/40	S4	1/4"-1/8" Gas-NPT	AISI316L	AISI316L	—	AISI316TIG
63	S10/3	≤ 40	N1	1/4" Gas-NPT	OT58brass	phosphor bronze	—	Tin alloy
63	S10/3	60/400	S1	1/4" Gas-NPT	OT58brass	—	phosphor bronze	Silver alloy
100	S10/3	≤ 40	N1	1/4" NPT, 1/2" Gas-NPT	OT58brass	phosphor bronze	—	Tin alloy
100	S10/3	60	S2	1/4" NPT, 1/2" Gas-NPT	OT58brass	AISI316L	—	Silver alloy
100	S10/3	100/1000	S2	1/4" NPT, 1/2" Gas-NPT	OT58brass	—	AISI316L	Silver alloy
63	S18/3	≤ 40	S4	1/4" Gas-NPT	AISI316	AISI316L	—	AISI316TIG
63	S18/3	60/400	S4	1/4" Gas-NPT	AISI316	—	AISI316L	AISI316TIG
100 ÷ 150	S18/3	≤ 60	S4	1/4" NPT, 1/2" Gas-NPT	AISI316	AISI316L	—	AISI316TIG
100 ÷ 150	S18/3	100/1000	S4	1/4" NPT, 1/2" Gas-NPT	AISI316	—	AISI316L	AISI316TIG

DS 100

S10/3 - MOUNTING STYLES AND DIMENSIONS



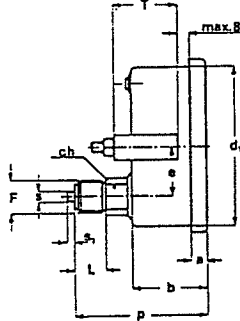
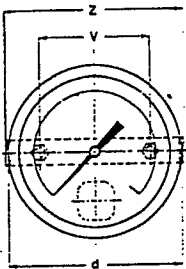
TYPE A
Lower Connection - Stem Mounting.



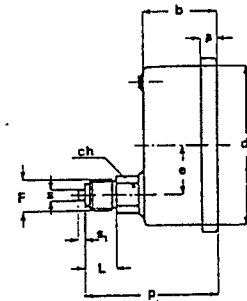
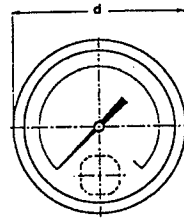
TYPE C
Lower Connection - Surface Mounting Back Flange.

TAB. 16 - DIMENSIONS AND WEIGHTS TYPE A-C

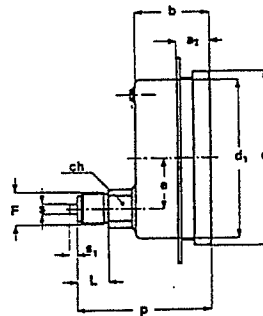
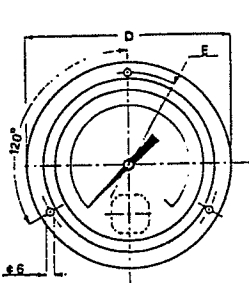
F	a	b	b ₁	c	c ₁	d	d ₁	D	E	h	h ₁	L	ch	Kg
1/2" Gas / NPT	10	47	51	16	20	110	101	130	116	86	52	20	□ 24	≈ 0.900
1/2" NPT	10	47	51	16	20	110	101	130	116	81	52	15	□ 24	



TYPE B
Back Connection - Flush Mounting, U-Clamp.



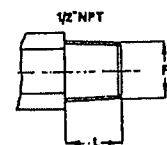
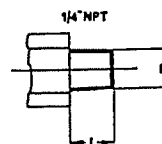
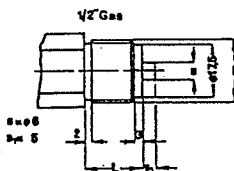
TYPE D
Back Connection - Stem Mounting.



TYPE E
Back Connection - Flush Mounting Front Flange.

TAB. 17 - DIMENSIONS AND WEIGHTS TYPE B-D-E

F	a	a ₂	b	d	d ₁	D	E	e	L	p	T	V	Z	ch	Kg.
1/2" Gas / NPT	10	18	47	110	101	130	116	31	20	86	41,5	70	112	□ 24	0,900
1/4" NPT	10	18	47	110	101	130	116	31	15	81	41,5	70	112	□ 24	

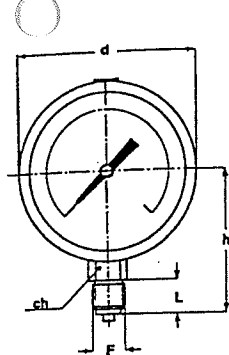


S10/3

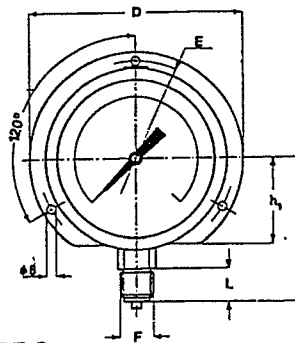
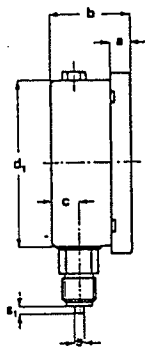
S18/3

DS 100 AND 150

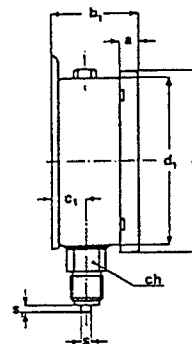
S18/3 - MOUNTING STYLES AND DIMENSIONS



TYPE A
Lower Connection - Stem Mounting.

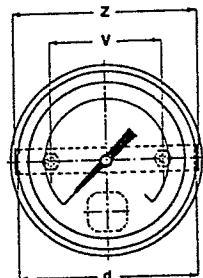


TYPE C
Lower Connection - Surface Mounting Back Flange.

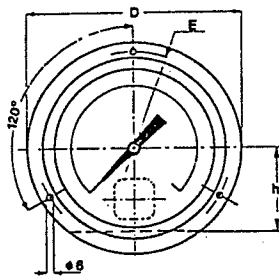
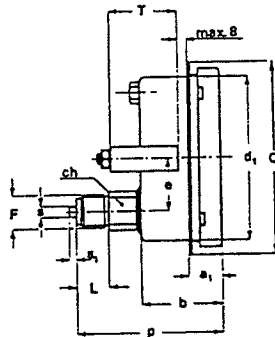


TAB.18-DIMENSIONS AND WEIGHT TYPE A-C

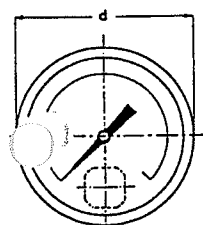
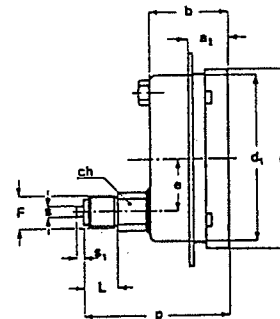
DS	F	a	b	b ₁	c	c ₁	d	d ₁	D	E	h	h ₁	L	ch	Kg
100	1/4" NPT	13	48	52	16	20	110,5	101	130	116	81	52	15	□ 24	≈ 1.000
100	1/2" Gas/NPT	13	48	52	16	20	110,5	101	130	116	86	52	20	□ 24	≈ 1.000
150	1/4" NPT	15	50,5	54	16,5	20	161,5	150	190	175	113	85	15	□ 24	≈ 1.800
150	1/2" Gas/NPT	15	50,5	54	16,5	20	161,5	150	190	175	118	85	20	□ 24	≈ 1.800



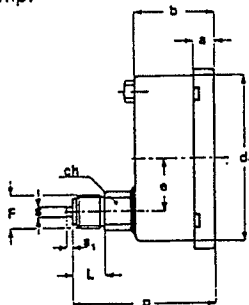
TYPE B
Back Connection -
Flush Mounting, U-Clamp.



TYPE E
Back Connection - Flush Mounting Front Flange.

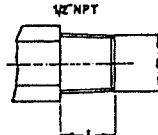
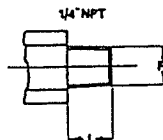
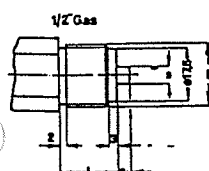


TYPE D
Back Connection - Stem Mounting.

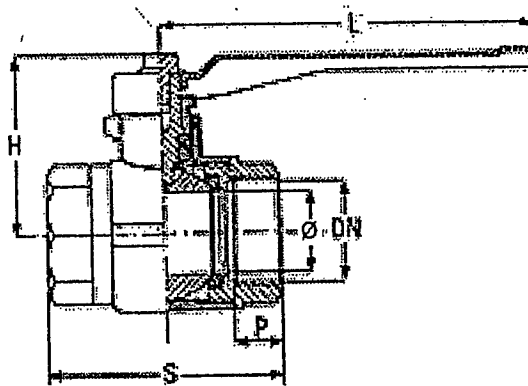
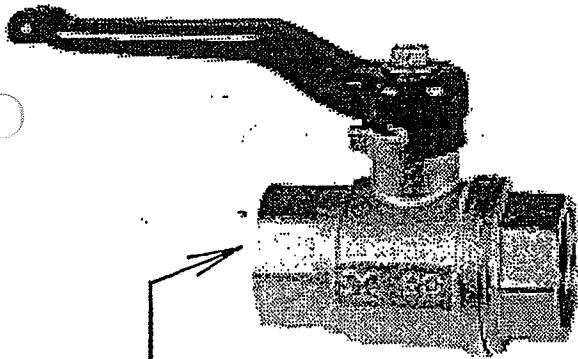


TAB.19-DIMENSIONS AND WEIGHT TYPE B-D-E

DS	F	a	a ₁	a ₂	b	d	d ₁	D	E	e	L	p	T	V	Z	Q	ch	Kg
100	1/4" NPT	13	20	20	48	110,5	101	130	116	31	15	82	41,5	70	112	112	□ 24	≈ 1.000
100	1/2" Gas/NPT	13	20	20	48	110,5	101	130	116	31	20	87	41,5	70	112	112	□ 24	≈ 1.000
150	1/4" NPT	15	21	25	50,5	161,5	150	190	175	48	15	80	41,5	106	155	164	□ 24	≈ 1.800
150	1/2" Gas/NPT	15	21	25	50,5	161,5	150	190	175	48	20	85	41,5	106	155	164	□ 24	≈ 1.800



5.1



Ball Valve PN 40 with lever

*both ends female threads full flow
brass nickel-plated
Type 105*

Ball Valve PN 40 with lever

*female/male thread full flow
brass nickel-plated
Type 120*

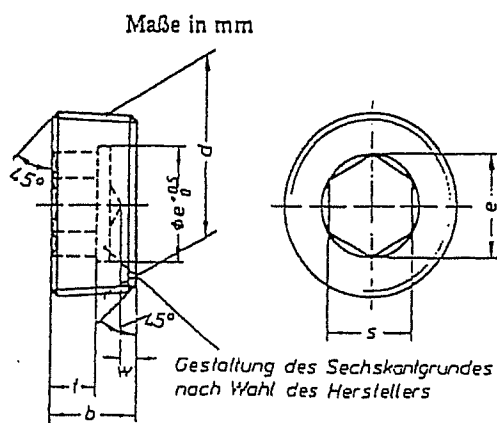
DN Zoll	ø	S	L	H	P	
1/4"	6	46	92	47	12	90 025 01
3/8"	10	46	92	47	12	90 025 02
1/2"	15	56	92	50	12	90 025 03
3/4"	20	66	92	55	16	90 025 04
1"	25	76	117	61	16	90 025 05
1 1/4"	32	86	150	78	17	90 025 06
1 1/2"	39	97	150	82	20	90 025 07
2"	48	112	150	88	20	90 025 08
2 1/2"	64	145	195	110	25	90 025 09
3"	78	157	195	120	27	90 025 19

ø	S	L	H	P	
6	45	92	47	10	90.025 10
10	45	92	47	12	90 025 11
15	55	92	50	12	90 025 12
20	67	92	55	16	90 025 13
25	75	117	61	16	90 025 14
32	88	150	78	17	90 025 15
39	100	150	82	20	90 025 16
48	112	150	88	20	90 025 17

5.1.1

Hexagonal head screw plug
Taper thread

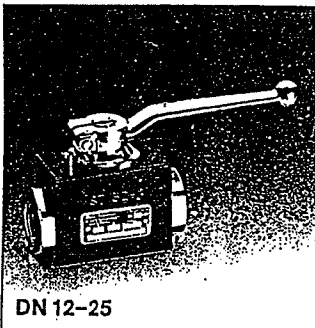
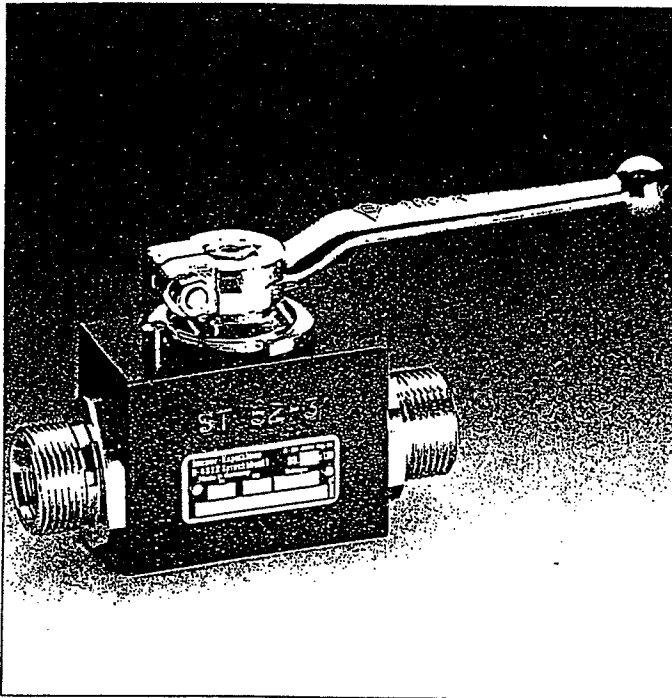
DIN
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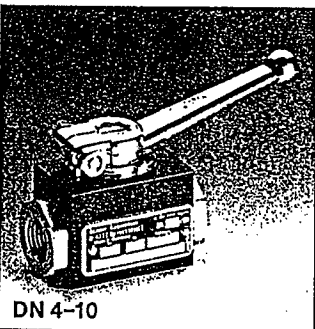
G 3/4

BHG

Block-Hochdruck-Kugelhahn mit Gewindeanschluß
Block type high pressure ball valve with threaded ports
Robinet monobloc haute pression à boisseau sphérique, raccords mâles ou femelles



DN 12-25



DN 4-10

● Der Schaltgriff gehört zum Lieferumfang.

● Bei Kugelhähnen mit Dichtungen aus Teflon werden im Normalfall O-Ringe aus Viton eingebaut.

● Sonderausführungen auf Anfrage.

● The handle is included.

● Ball valves with teflon seals are normally delivered with viton O-rings.

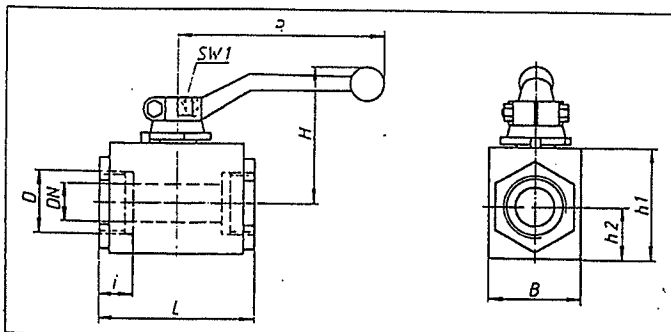
● Special designs upon request.

● La manette est comprise dans la livraison.

● Les robinets à boisseau sphérique avec joints teflon, sont prévus normalement avec joints torique en viton.

● Exécutions spéciales sur demande.

	PN bar	DN mm	Normalausführungen Standard versions Exécutions standards	
			Typ 1	Typ 2
Gehäuse Body Corps	150-500	4-25	Stahl Steel Acier	Stahl Steel Acier
Kugel Ball Bille	150-500	4-25	Stahl, hartverchromt Steel, hard chromed Acier, chromé dur	Stahl, hartverchromt Steel, hard chromed Acier, chromé dur
Dichtungen Seals Joints	150-500	4-25	Polyamid	Teflon



Bestell-Nr. und Typ
bitte angeben.

Please quote order-no.
and typ.

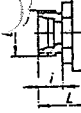
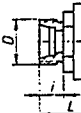
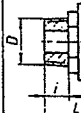
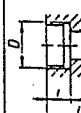
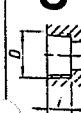
Mentionner référence
et type s.v.p.

Typ 1
Typ 2
400
BHG V500.104/...

Volldurchgang
Baulänge DIN 3202 V1, M3
DN 4- 25
PN 150-500

Full bore
Length, face-to-face DIN 3202 V1, M3
DN 4- 25
PN 150-500

Passage intégral
Longueur hors tout DIN 3202 V1, M3
DN 4- 25
PN 150-500

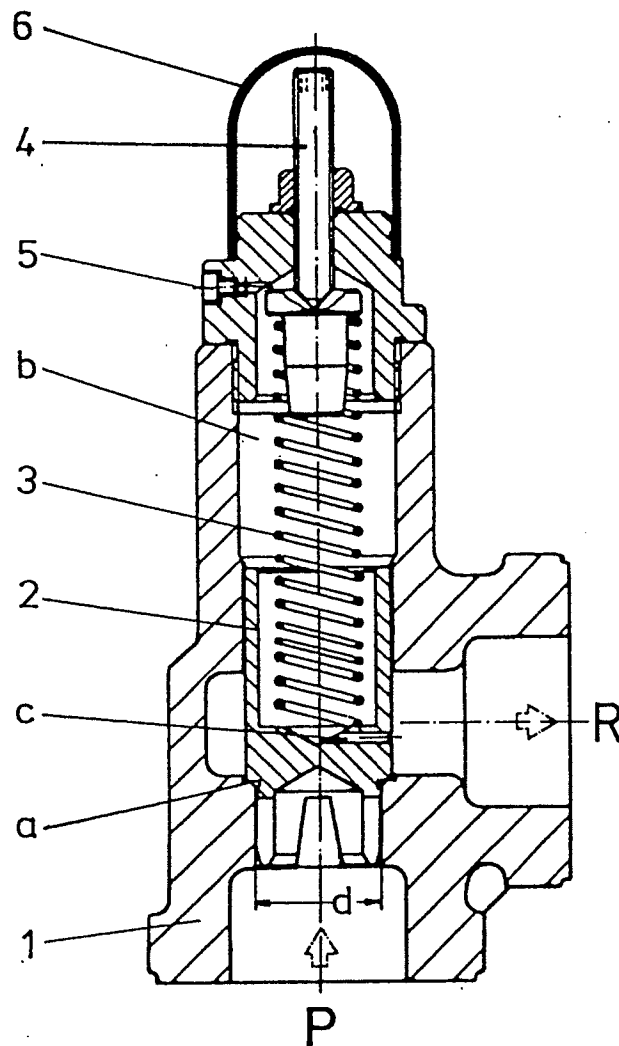
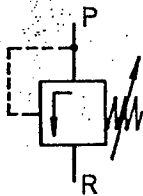
Anschlußart Type of connection Type de raccordement	DN	Rohr AD pipe OD ø ext. du tube	PN		D	i	L	B	h ₁	h ₂	H	R	SW1	Gew. kg Weight kg Poids kg	Bestell-Nr. Order-No. Référence
			Poly- amid	Teflon											
1  Für Rohrverschraubung mit Kugelbuchse, DIN 7601 For compression fitting, with ball-type nipple, DIN 7601 Pour raccord avec cône, DIN 7601	4	-	500	350	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.104
	6	-	500	350	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.106
	8	-	500	350	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 500.108
	10	-	500	350	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 500.110
	12	-	500	350	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 500.112
	16	-	400	250	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 400.116
	20	-	315	175	M 30x1,5	14	110	45	55	24	77	180	10	1,3	BHG V 315.120
	25	-	315	150	M 38x1,5	14	120	55	65	30,5	82	180	12	2,2	BHG V 315.125
2  Für Rohrverschraubung leichte Reihe, DIN 2353 For compression fitting, light series, DIN 2353 Pour raccord avec cône DIN 2353, série légère	4	6	250*	250*	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 250.204
	6	8	250*	250*	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 250.206
	8	10	250*	250*	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 250.208
	10	12	250*	250*	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 250.210
	12	15	250*	250*	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 250.212
	16	18	160*	160*	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 160.216
	20	22	160*	160*	M 30x2	14	110	45	55	24	77	180	10	1,3	BHG V 160.220
	25	28	100*	100*	M 36x2	14	120	55	65	30,5	82	180	12	2,2	BHG V 100.225
3  Für Rohrverschraubung schwere Reihe, DIN 2353 For compression fitting, heavy series, DIN 2353 Pour raccord avec cône DIN 2353, série lourde	4	8	500	350	M 16x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.304
	6	10	500	350	M 18x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.306
	8	12	500	350	M 20x1,5	12	76	25	30	12	45	90	8	0,3	BHG V 500.308
	10	14	500	350	M 22x1,5	14	80	30	35	15	51	90	8	0,4	BHG V 500.310
	12	16	400*	350	M 24x1,5	14	96	35	45	18	66	130	10	0,8	BHG V 400.312
	16	20	400	250	M 30x2	16	105	40	50	21,5	68	130	10	1,0	BHG V 400.316
	20	25	315	175	M 36x2	18	110	45	55	24	77	180	10	1,3	BHG V 315.320
	25	30	250*	150	M 42x2	20	120	55	65	30,5	82	180	12	2,2	BHG V 250.325
4  Flachdichtend, metrisches Außengewinde, DIN 13 Flat sealing metric male thread, DIN 13 Filet métrique mâle avec étanchéité, DIN 13	4	-	500	350	M 12x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.404
	6	-	500	350	M 14x1,5	10	76	25	30	12	45	90	8	0,3	BHG V 500.406
	8	-	500	350	M 16x1,5	11	76	25	30	12	45	90	8	0,3	BHG V 500.408
	10	-	500	350	M 18x1,5	11	80	30	35	15	51	90	8	0,4	BHG V 500.410
	12	-	500	350	M 22x1,5	12	96	35	45	18	66	130	10	0,8	BHG V 500.412
	16	-	400	250	M 26x1,5	12	105	40	50	21,5	68	130	10	1,0	BHG V 400.416
	20	-	315	175	M 30x1,5	14	110	45	55	24	77	180	10	1,3	BHG V 315.420
	25	-	315	150	M 38x1,5	14	120	55	65	30,5	82	180	12	2,2	BHG V 315.425
5  Flachdichtend, Rohraußen- gewinde, DIN ISO 228/1 Flat sealing BSP male thread, DIN ISO 228/1 Filet whitworth mâle avec étanchéité, DIN ISO 228/1	4	-	500	350	G 1/8 A	10	76	25	30	12	45	90	8	0,3	BHG V 500.504
	6	-	500	350	G 1/4 A	11	76	25	30	12	45	90	8	0,3	BHG V 500.506
	8	-	500	350	G 3/8 A	12	76	25	30	12	45	90	8	0,3	BHG V 500.508
	10	-	500	350	G 1/2 A	12	80	30	35	15	51	90	8	0,4	BHG V 500.510
	12	-	500	350	G 3/4 A	12	96	35	45	18	66	130	10	0,8	BHG V 500.512
	16	-	400	250	G 1 A	14	105	40	50	21,5	68	130	10	1,0	BHG V 400.516
	20	-	315	175	G 1 1/4	16	110	45	55	24	77	180	10	1,3	BHG V 315.520
	25	-	315	150	G 1 1/2 A	18	120	55	65	30,5	82	180	12	2,2	BHG V 315.525
6  NPT-Außengewinde, ANSI/ASME B 1.20.1 NPT male thread, ANSI/ASME B 1.20.1 Filet mâle NPT, ANSI/ASME B 1.20.1	4	-	500	350	NPT 1/8	9,5	69	25	30	12	45	90	8	0,3	BHG V 500.604
	6	-	500	350	NPT 1/4	14,5	79	25	30	12	45	90	8	0,3	BHG V 500.606
	8	-	500	350	NPT 3/8	14,5	79	25	30	12	45	90	8	0,3	BHG V 500.608
	10	-	500	350	NPT 1/2	19	90	30	35	15	51	90	8	0,4	BHG V 500.610
	12	-	500	350	NPT 3/4	19	110	35	45	18	66	130	10	0,8	BHG V 500.612
	16	-	400	250	NPT 1	19	112	40	50	21,5	68	130	10	1,0	BHG V 400.616
	20	-	315	175	NPT 1 1/4	24	125	45	55	24	77	180	10	1,4	BHG V 315.620
	25	-	315	150	NPT 1 1/2	24,5	141	55	65	30,5	82	180	12	2,3	BHG V 315.625
7  Rohrrinnengewinde, DIN ISO 228/1 BSP female thread, DIN ISO 228/1 Taraudage, whitworth DIN ISO 228/1	4	-	500	350	G 1/8	11	50	25	30	12	45	90	8	0,3	BHG V 500.704
	6	-	500	350	G 1/4	12,5	50	25	30	12	45	90	8	0,3	BHG V 500.706
	8	-	500	350	G 3/8	12,5	70	25	30	12	45	90	8	0,3	BHG V 500.708
	10	-	500	350	G 1/2	12,5	60	30	35	15	51	90	8	0,4	BHG V 500.710
	12	-	500	350	G 3/4	15	75	35	45	18	66	130	10	0,7	BHG V 500.712
	16	-	400	250	G 1	15	75	40	50	21,5	68	130	10	0,9	BHG V 400.716
	20	-	315	175	G 1 1/4	18	80	45	55	24	77	180	10	1,2	BHG V 315.720
	25	-	315	150	G 1 1/2	20	90	55	65	30,5	82	180	12	2,1	BHG V 315.725
8  NPT-Innengewinde, ANSI/ASME B 1.20.1 NPT female thread, ANSI/ASME B 1.20.1 Taraudage NPT, ANSI/ASME B 1.20.1	4	-	500	350	NPT 1/8	11	50	25	30	12	45	90	8	0,3	BHG V 500.804
	6	-	500	350	NPT 1/4	12,5	50	25	30	12	45	90	8	0,3	BHG V 500.806
	8	-	500	350	NPT 3/8	12,5	70	25	30	12	45	90	8	0,3	BHG V 500.808
	10	-	500	350	NPT 1/2	12,5	60	30	35	15	51	90	8	0,4	BHG V 500.810
	12	-	500	350	NPT 3/4	15	75	35	45	18	66	130	10	0,7	BHG V 500.812
	16	-	400	250	NPT 1	15	75	40	50	21,5	68	130	10	0,9	BHG V 400.816
	20	-	315	175	NPT 1 1/4	18	80	45	55	24	77	180	10	1,2	BHG V 315.820
	25	-	315	150	NPT 1 1/2	20	90	55	65	30,5	82	180	12	2,1	BHG V 315.825

* Druckstufe der Rohrverschraubung.
Do not exceed nominal pipe pressure. Veuillez vous reporter à la pression nominale du tube.

Aufbau der Druckbegrenzungsventile SPV, SPVF

Construction of the Pressure Relief Valves SPV, SPVF

Sinnbild Symbol



- 1 Gehäuse
- 2 Ventilschieber
- 3 Druckfeder
- 4 Einstellschraube
- 5 Entlüftungsschraube
- 6 Schutzkappe

- 1 Housing
- 2 Sliding Piston
- 3 Compression Spring
- 4 Set Screw
- 5 Bleeding Screw
- 6 Protective Cap

Beschreibung

Das Druckbegrenzungsventil SPV/SPVF ist ein direkt-gesteuertes Schieberventil für Einbau in Rohrleitungen und dient zur Absicherung von Niederdruck-Hydraulik-Kreisläufen bis $p_{\max}$ 20 bar (30 bar). Der Leitungs-Anschluß kann mittels SAE-Flansch (3000 psi) oder Whitworth-Rohrgewinde (G) vorgenommen werden.

Ventilaufbau:

Der Ventilschieber 2 wird durch die Druckfeder 3 gegen die Ringfläche a gedrückt und sperrt somit über den Durchmesser d den Pumpenanschluß P vom Tankanschluß R ab. Bei Erreichen des Betriebsdruckes p, eingestellt durch die Einstellschraube 4, gibt der Ventilschieber 2 den Ölstrom zum Tank frei. Der Federraum b wird durch die Bohrung c entlastet.

Bei Inbetriebnahme des Ventils ist der Federraum b durch die Entlüftungsschraube 5 (SW 4) zu entlüften. Das Ventil sollte vorzugsweise senkrecht, mit der Einstellschraube nach unten, montiert werden. Die Ventile sind in vier verschiedenen Druckbereichen lieferbar; da jede Druckfeder infolge ihrer Charakteristik nur für einen begrenzten Einstellbereich geeignet ist.

Description

The direct spring, sliding piston Pressure Relief Valve SPV/SPVF is intended for inline mounting and is suitable to safeguard low pressure hydraulic circuits up to 20 (30) bars. The pipe connection is to be effected either by SAE-Mounting Surfaces (3000 psi) or by Whitworth Pipe Threads "G".

Valve Construction:

The Sliding Piston 2 is pressed against the Annulated Area a by the Compression Spring 3 and separates the Pump Connection P from the Reservoir Connection R sealing the Bore d. As soon as the operating pressure p adjusted by the Set Screw 4, is reached the Sliding Piston 2 releases the oil flow to the reservoir. The Spring Chamber b is pressure compensated by the Bore c.

During starting-up the valve the Spring Chamber b must be bled by the Bleeding Screw 5, (Hex. Socket Width: 4). The valve should preferably be installed in vertical fitting position with the pressure setting arrangement down.

The pressure relief valves are available in four different pressure setting ranges because, due to their spring-rates, each of the compression springs can only cover a limited pressure setting range.

Kenngrößen

Characteristics

Kenngrößen nach VDI 3267

Druckbegrenzungsventil

direktgesteuert

Nenngröße:	10 25 40 50 80
max. Durchfluß:	40 90 450 550 800 l/min
Einbaudruck:	30 20 20 20 20 bar

Allgemeine Kenngrößen

Bauart:	Schieberventil
Befestigungsart:	in Rohrleitung
Leitungsanschluß:	SAE-Flansch (3000 psi) Whitworth-Gewinde G
Geräteabmessungen:	Seite 7, 9, 10
Gewicht:	Seite 7, 9, 10
Einbaulage:	Druckeinstellschraube unten

Umgebungstemperatur:	$\vartheta_{u \min} = -20^{\circ}\text{C}$ $\vartheta_{u \max} = +60^{\circ}\text{C}$
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Hydraulische Kenngrößen

	NG 10 25 40 50 80
Einstelldruckbereich:	$p_{v \min}$ 0,5 0,5 0,5 0,5 0,5 bar $p_{v \max}$ 30 20 20 20 20 bar
Druckmitteltemperatur:	$\vartheta_{m \min} = -20^{\circ}\text{C}$ $\vartheta_{m \max} = +80^{\circ}\text{C}$
Viskosität:	$\nu_{\min} = 6 \text{ mm}^2/\text{s}$ $\nu_{\max} = 600 \text{ mm}^2/\text{s}$
Δp -Q-Kennlinie:	Seite 5
Druckflüssigkeiten:	Hydrauliköle nach DIN 51524/25 Andere Flüssigkeiten auf Anfrage

Betätigungsart

mechanisch	Einstellschraube Drehgriff
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Zubehör

Anschweißflansch SAE (3000 psi)	Seite 10
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Characteristics acc. to VDI 3267

Pressure Relief Valve

directly spring operated

Nom. Size:	10 25 40 50 80
Max. Flow Capacity:	40 90 450 550 800 l/min
Nom. Operating Pressure:	30 20 20 20 20 bars

General Characteristics

Construction:	Sliding Piston Relief Valve
Mounting:	Inline
Pipe Connection:	SAE-Flange (3000 psi) Whitworth Pipe Thread "G"
Dimensions:	Pages 7, 9, 10
Weight:	Pages 7, 9, 10
Fitting Position:	Set Screw down

Perm. Ambient Temp.:	$\vartheta_{u \min} = -20^{\circ}\text{C}$ $\vartheta_{u \max} = +60^{\circ}\text{C}$
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Hydraulic Characteristics

	NS 10 25 40 50 80
Pressure Setting Ranges:	$p_{v \min}$ 0,5 0,5 0,5 0,5 0,5 bars $p_{v \max}$ 30 20 20 20 20 bars
Fluid Temperature Range:	$\vartheta_{m \min} = -20^{\circ}\text{C}$ $\vartheta_{m \max} = +80^{\circ}\text{C}$
Viscosity Range:	$\nu_{\min} = 6 \text{ mm}^2/\text{s}$ $\nu_{\max} = 600 \text{ mm}^2/\text{s}$
Δp -Q-Characteristics:	Page 5
Hydraulic Fluids:	Hydraulic Oils acc. to DIN 51524/25 other fluids on request

Type Setting

mechanical	Set Screw Knob
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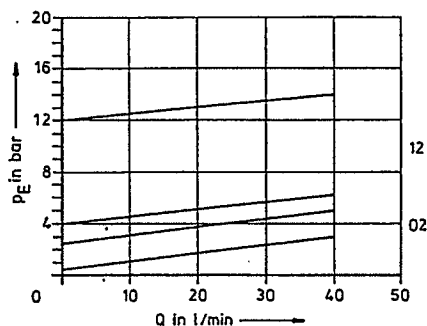
Accessories

Welding Flange SAE (3000 psi)	Page 10
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Kennlinie

$p_E - Q$

Viskosität = 34 mm²/s



02
12
30

Characteristic Curves

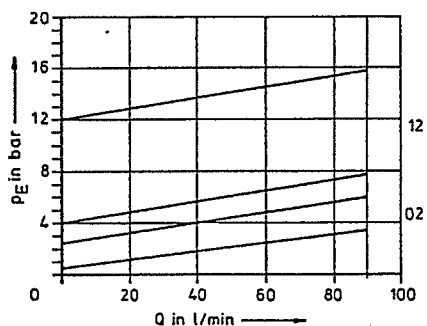
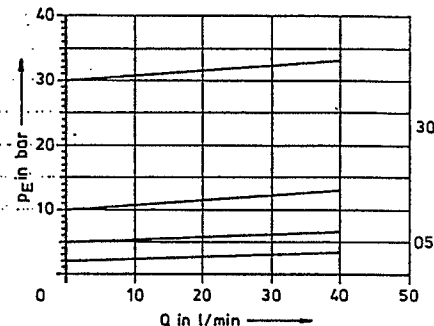
$p_E - Q$

Viscosity = 34 mm²/s

SPV 10

Druckbereiche
Pressure Setting Ranges

05
30

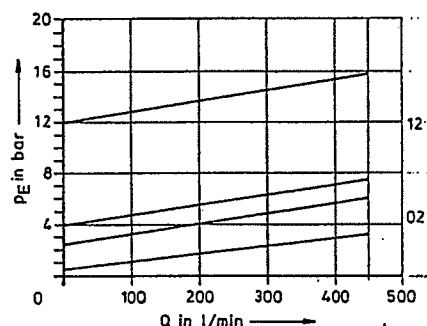
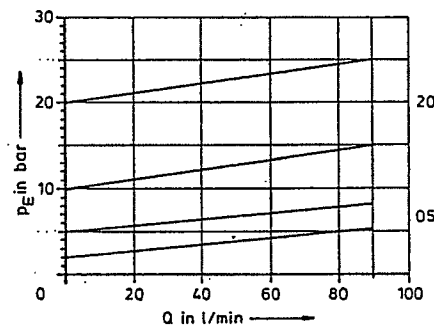


02
12

SPVF 25

Druckbereiche
Pressure Setting Ranges

05
20

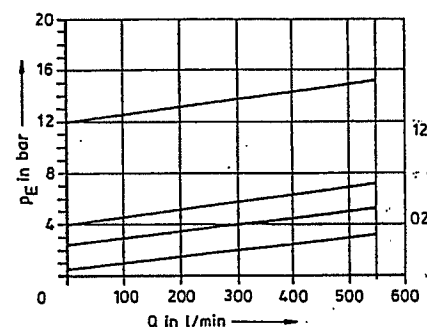
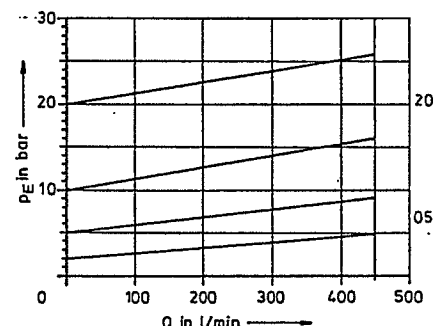


02
12

SPVF 40

Druckbereiche
Pressure Setting Ranges

05
20

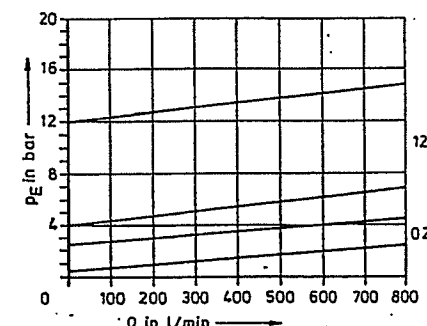
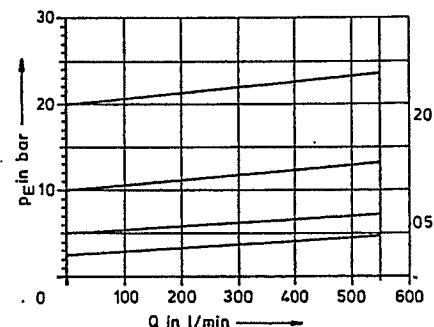


02
12

SPVF 50

Druckbereiche
Pressure Setting Ranges

05
20

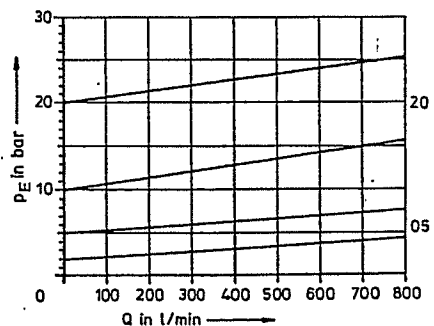


02
12

SPVF 80

Druckbereiche
Pressure Setting Ranges

05
20



Typenschlüssel Type Key

Nenngröße 10

Nominal Size 10

Druckbegrenzungsventil
direktgesteuert

Pressure Relief Valve
directly spring operated

SPV . 10 . 1G 1 . .

Gerätetyp
Valve Type

SPV

Pulteinbau
Panel Mounting

M

Nenngröße
Nom. Size

10

Q_{max} 40 l/min

Ausführungsart:
Models:

Standard-Ausführung
Standard-Type

A

Hochtemperatur-Ausführung
bis + 220°C
High Temperature-Type
up to + 220°C

B

FPM-Ausführung bis 150°C
FPM-Type up to 150°C

C

Leitungsanschluß:
Pipe Connection:

mit Gewindeanschluß G 1/2
by Threaded Ports G 1/2

1G

Konstruktions-Kennziffer:
Design Serial No.:

Interne Vergabe
Specified by KRACHT

1

Druckbereich:
Pressure Setting Ranges:

02	0,5 ... 2,5 bar	bars
05	2 ... 5 bar	bars
12	4 ... 12 bar	bars
30	10 ... 30 bar	bars

Betätigungsart:
Type of Pressure Setting:

A	Einstellschraube Set Screw
B	Drehgriff Knob

Bestellbeispiel: **SPV 10 B 1G 1 A12**

Druckbegrenzungsventil direktgesteuert, Nenngröße 10
(Q_{max} 40 l/min), Hochtemperatur-Ausführung (... + 220°C)
mit Gewindeanschluß G 1/2, mit Einstellschraube,
Druckbereich 4 ... 12 bar

Ordering Example: **SPV 10 B 1G 1 A12**

Pressure Relief Valve, directly spring operated, Nominal Size 10
(Q_{max} 40 l/min), High Temperature-Type (up to + 220°C)
with Threaded Ports G 1/2, Pressure Setting by Set Screw,
Pressure Setting Range: 4 up to 12 bars

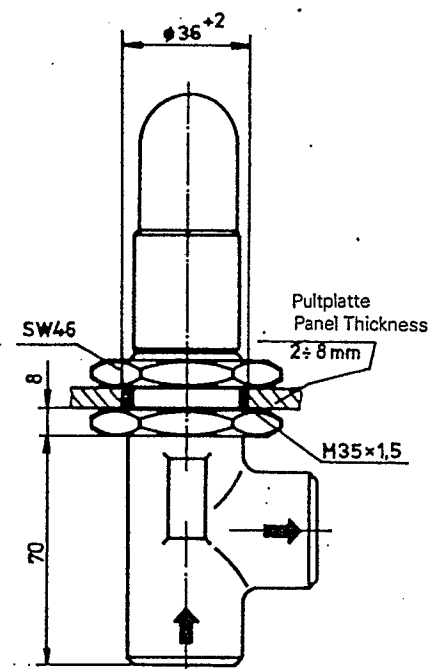
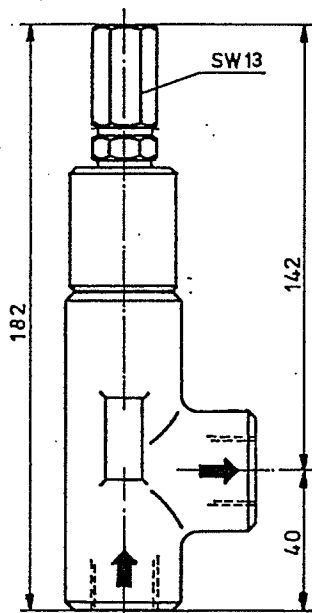
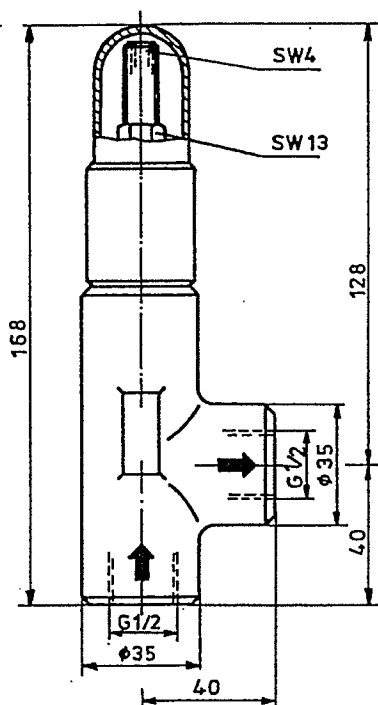
Bestellbeispiel: **SPVM 10 A 1G 1 B30**

Druckbegrenzungsventil direktgesteuert für Pulteinbau,
Nenngröße 10 (Q_{max} 40 l/min), Standard-Ausführung,
mit Gewindeanschluß G 1/2, mit Drehgriff, Druckbereich
10 ... 30 bar

Ordering Example: **SPVM 10 A 1G 1 B30**

Pressure Relief Valve, directly spring operated for
Panel Mounting, Nominal Size 10 (Q_{max} 40 l/min), Standard-Type
with Threaded Ports G 1/2, Pressure Setting by Knob,
Pressure Setting Range: 10 up to 30 bars

Geräteabmessungen Dimensions



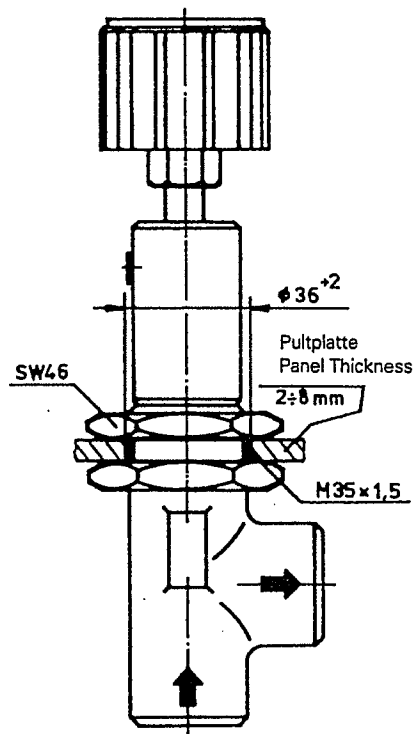
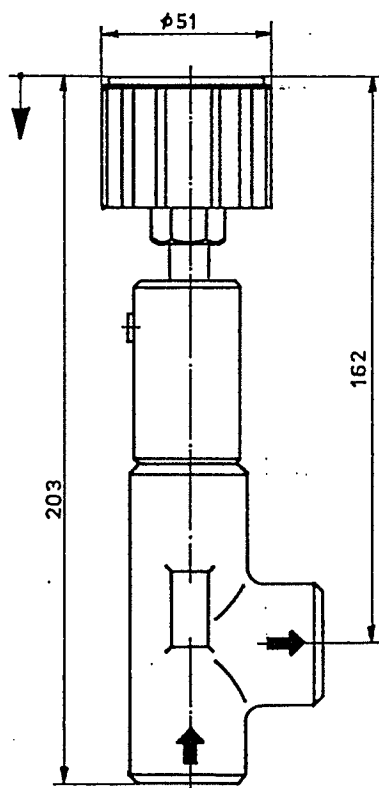
Bestellbezeichnung Ordering Code

SPV 10 A 1G 1 A .

SPV 10 B 1G 1 A .

SPVM 10 A 1G 1 A .

Verstellhub
max. 26 mm
Adjusting
Travel to max.
pressure setting:
max. 26 mm



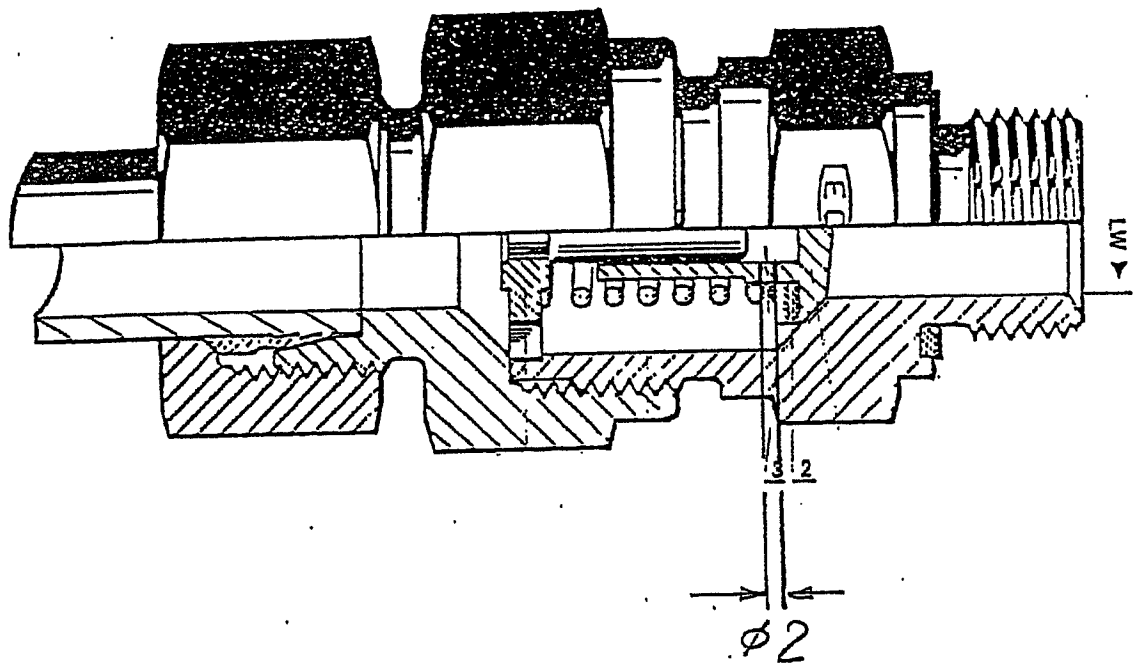
Gewicht 0,9 kg
Weight 0,9 kg

SW = Wrench Size
or
Hex. Socket Width

Bestellbezeichnung Ordering Code

SPV 10 A 1G 1 B .

SPVM 10 A 1G 1 B .



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INTEGRAL HYDRAULIK Anlagen + Systeme												
Werkstoff X												
Halbzeug-Nr. X												
Allgemeintoleranzen ISO 2768-m über 0,5 6 30 120 400 bis 6 30 120 400 1000 ±0,1 ±0,2 ±0,3 ±0,5 ±0,8			Ersatz für	Zeichen			Änderungen			Ä.-M.-Nr	Datum	Nr
			Ersetzt durch	gez.	24.09.03	Name	Zeichnungs Nr.					
Oberflächenangaben nach DIN ISO 1302				N. gepr.			016-1531-014-014					
ID-Nr.			Gewicht	Maßstab	Benennung							
					Check valve							

IN LINE NEEDLE VALVES CHECK VALVES IN CARBON STEEL

FT 257/2

18.

A ~~X~~ B

In line double-acting flow control and shut-off control needle valves

They allow flow control in both directions. Needle adjustments give:

- full shut off (via metal seat);
- accurate control for a wide range of flowrates.

The valve has a graduated adjustment scale below the handle to indicate accurately the valve position.

There is a locking screw in the handle to allow the handle to be fixed. (Preventing accidental adjustment or movement due to vibration).

A panel mounting nut (G) can be supplied on request.

On request

- Versions in AISI 316 stainless steel code FT 2257/2
- Viton seals (V)
- NPT threads
- Complete with panel mounting nut (G)
- Handwheel in ABS (mp) - plastic

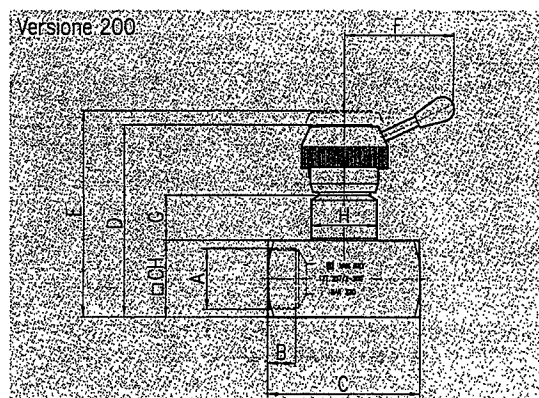
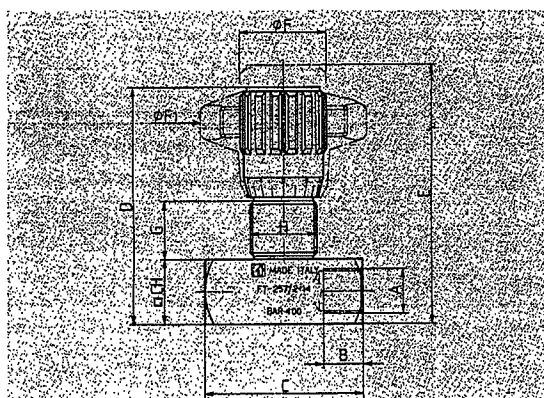


MATERIALS

Body	9 s Mn Pb 23 - UNI 5105
Needle	1 C 40 - UNI 8373
Or	Nitrile
Antiextrusion ring	PTFE
Handwheel	GD - Al Si 12 - UNI 5706 aluminium
Handwheel (mp)	ABS - plastic

EXAMPLE FOR ORDERING

Accessories on request					
	Code	Type	Panel ring	Viton seal	Handwheel in plastic
Steel	FT 257/2	18	G	V	mp
Stainl. steel	FT 2257/2	38	G	-	mp



CODE FT 257/2

Type	A UNI 338	B	C	D	E	F	F1	G	H	CH	Weight kg
18	1/8"G	8.5	38	59	64	22	40	13.5	M17x1	16	0.110
14	1/4"G	12.5	49	71	78	27	50	17	M20x1	20	0.200
38	3/8"G	12.5	59	84	93	33	70	19.5	M25x1.5	25	0.375
12	1/2"G	15.5	68	97	107	38	80	21	M30x1.5	30	0.600
34	3/4"G	17	86	120.5	132.5	47	100	26.5	M40x1.5	40	1.250
100	1"G	20	105	151.5	167.5	58	120	35	M50x1.5	50	2.550
114	1 1/4"G	22	120	156.5	172.5	58	120	35	M50x1.5	55	3.000
112	1 1/2"G	24	134	167	181	58	120	35	M55x2	65	4.217
200	2"G	27	150	188	202	85	/	44	M65x2	75	7.300

Note: Instructions for panel mounting on page 17

Specification

Type	air throughput [l/min]*			fig.	connection size ***	part number			
	Mic	Sm-L	Mol			Mic	Mic-UM	Sm-L	Sm-L-UM
Pi 0145	2100	1800	**	1	DIN	610.210.7	610.211.5	610.212.3	610.213.1
Pi 0146	1800	1500	**	1+1a	DIN	610.214.9	610.215.6	610.216.4	610.217.2
Pi 0147	2000	1700	**	3 **	40	610.218.0	610.219.8	610.220.6	610.221.4
Pi 0148	2400	2000	**	2 **	52	610.222.2	610.223.0	610.224.8	610.225.5
Pi 0149	1600	1300	**	4 **	1 1/2	610.226.3	610.227.1	610.228.9	610.229.7

* throughput at delta-p of 10 mbar

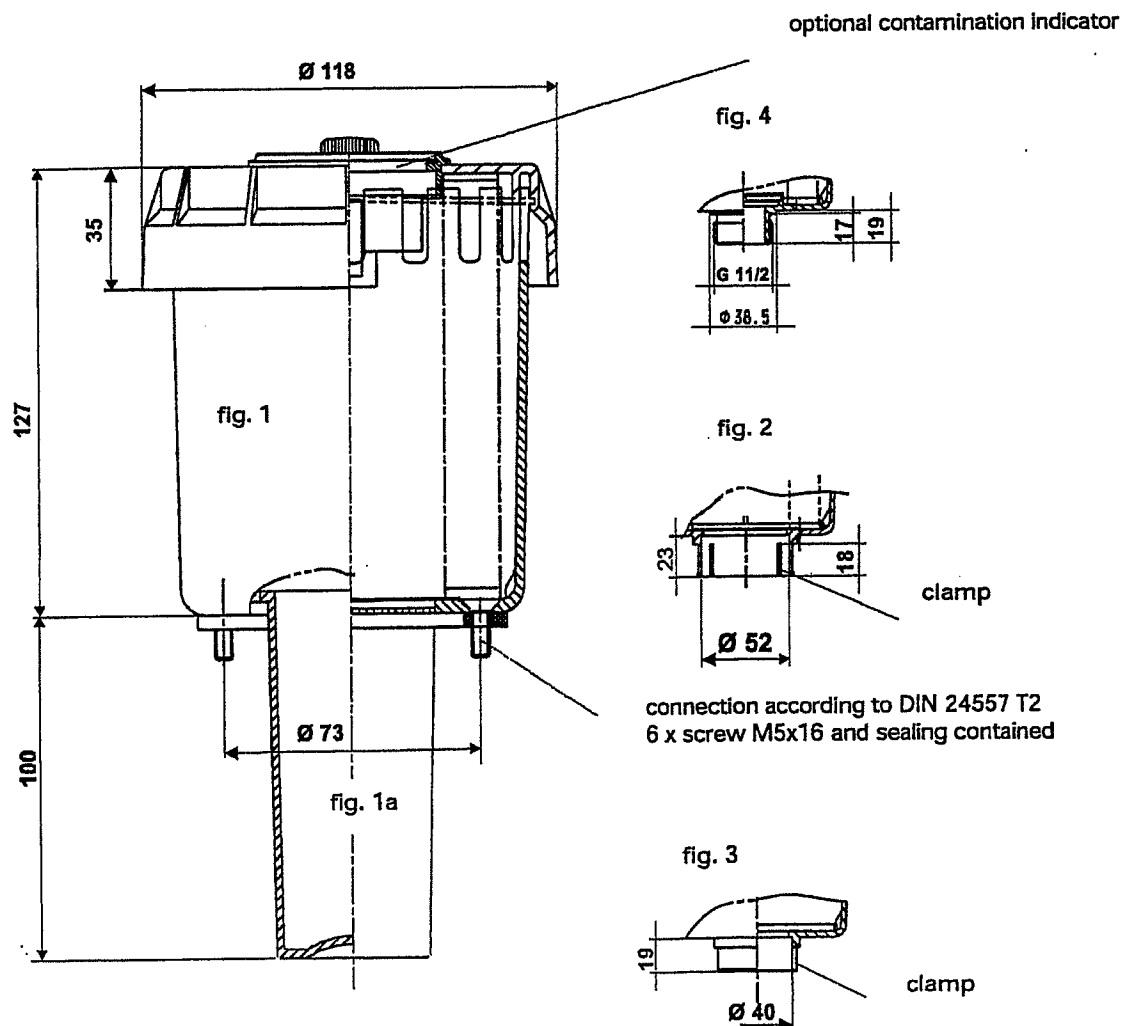
** in preparation

*** fit in the range H11

Retention rate	
Mic	10 µm > 99 %
Sm-L	3 µm > 99 %

Temperature range	-40 °C up to +100 °C
Housing material	black polyamide
Sealing material	rubberized cork
Compatibility	all hydraulic oils

Special variations on request



MAHLE Filtersysteme GmbH
 Bereich Industriefilter
 Schleifbachweg 45 D-74613 Öhringen
 P.O. Box 13 09 D-74603 Öhringen
 Tel./Phone +49 79 41 67-0
 Telefax +49 79 41 67-4 29
 Internet: www.mahle.com
 E-mail: ub2.Industrie@mahle.com

Subject to technical alteration!

1. Description

To economically operate today's modern hydraulic systems, the use of efficient, qualified filters is a must. Designers of such systems increasingly rely on consultancy from the product and application specialists of the component manufacturer. A broad experience and modern equipped laboratories, research and development departments are necessary to provide this assistance. Purolator offers this qualified service with each and every product.

The use of high grade materials and modern production methods guarantee maximum operating reliability. Before the release of a series, all filter housings undergo extensive pulsation and burst tests. The specified operating pressure can therefore be utilized. Special attention is given to a compact design, providing space-saving installation. The flow-promoting design of the components ensure a low pressure loss.

Purolator filter elements still set standards in filtration. Constant Multi-Pass-Test comparisons, carried out in accordance to the ISO 4572 Standards, confirm this.

The use of chemically resistant filter medias with a large absorption capability, for the multi-layer Super-micronic® elements and sturdy, asymmetric, impregnated cellulose materials for Micronic® elements, give the base for this success.

Filter surface, number and depth of pleats are optimized to provide a low differential pressure and the largest possible dirt holding capacity.

For servo or similar systems the use of Super-micronic® elements with high differential pressure resistance (Δp 210 bar) is recommended. To save costs and utilize all advantages of the Purolator filters, we recommend the use of contamination indicators. The differential pressure sensing signal is set to exploit the highest possible dirt holding capacity of the filter elements.

Contamination indicators with visual or visual/electrical indication – both, with "check-function" – are available. The electrical version features Normally Open and Normally Closed position and is also available with two-step-indication or cold start signal depression.

Replacement of the filter elements is easy, by simply removing the filter bowl.

- Modular design
- Easy to service
- Visual/electrical contamination indication
- World-wide sales activities

Operating pressure
25 (60) bars
Nominal flow rate
up to 450 l/min



**Purolator
Filter GmbH**

Unternehmensbereich Industriefilter

Postfach 1309 · Schleifbachweg 45

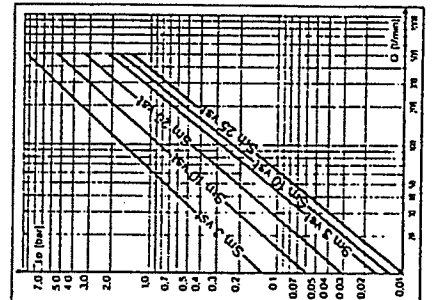
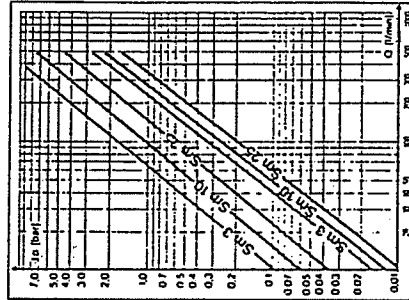
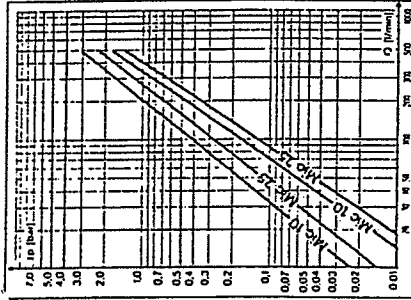
D-7110 Öhringen · Telefon (07941) 671 · Telex 74458

distributed by

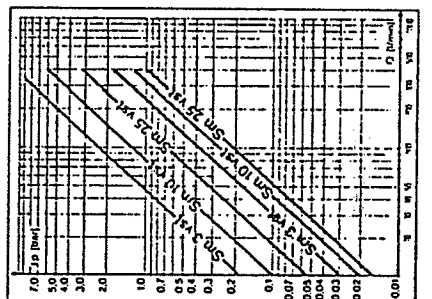
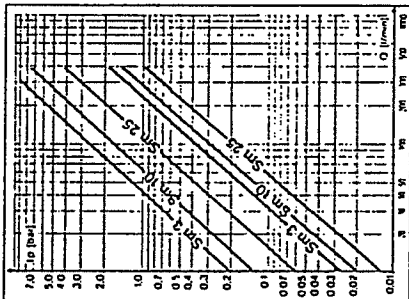
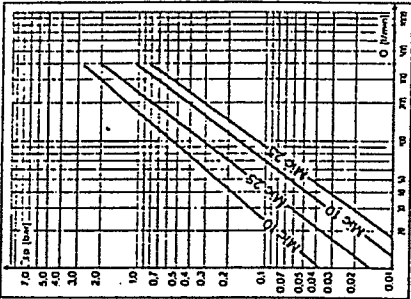
2. Flow rate/pressure drop curve compl. filter

190 mm²/s (25°E)
33 mm²/s (4,5°E)

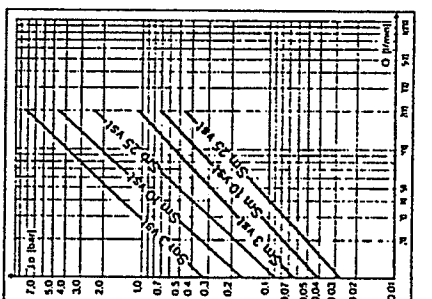
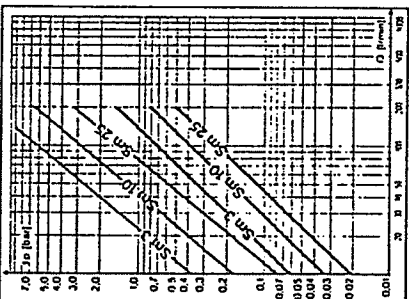
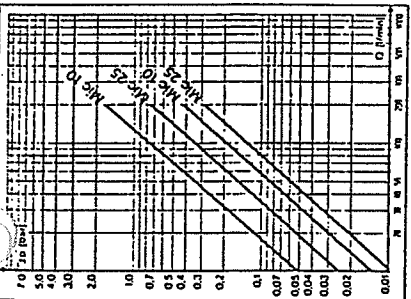
PI 2045..



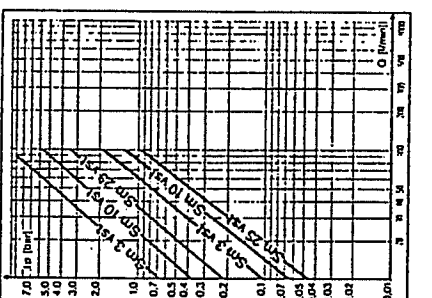
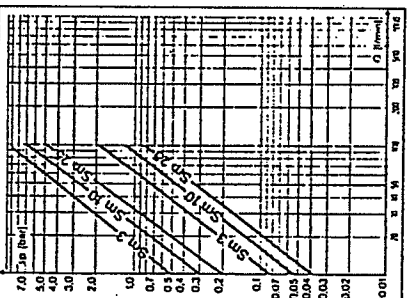
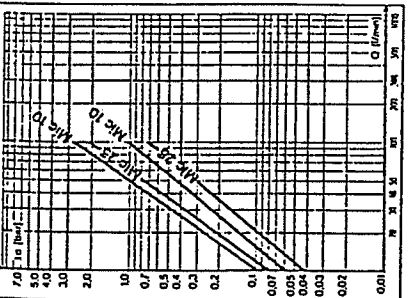
PI 2030..



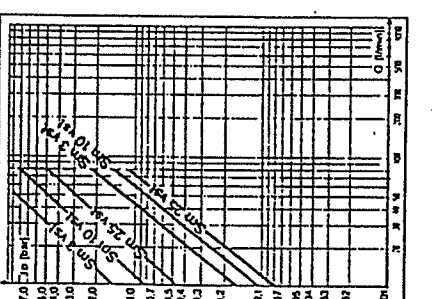
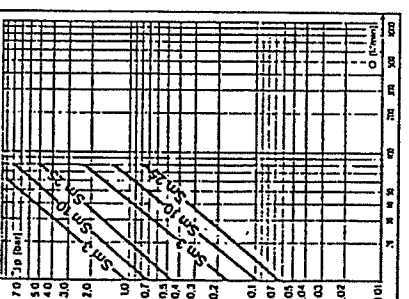
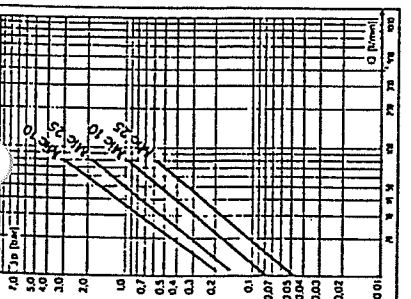
PI 2005..



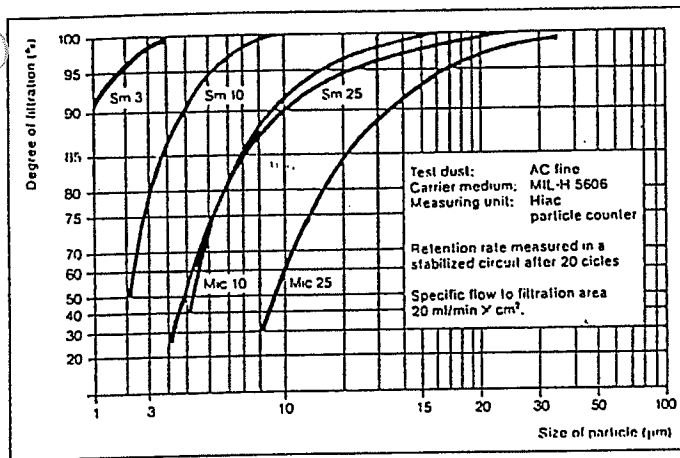
PI 2008..



PI 2005..



3. Separation characteristics



4. Filter performance data

tested according to ISO 4572 (Multi-Pass-Test)

Supermicronic™ elements
with Δp 20 bar

Sm 3 $\beta_1 \geq 75$
Sm 10 $\beta_1 \geq 75$
Sm 25 $\beta_{25} \geq 75$

at 7 bar differential pressure

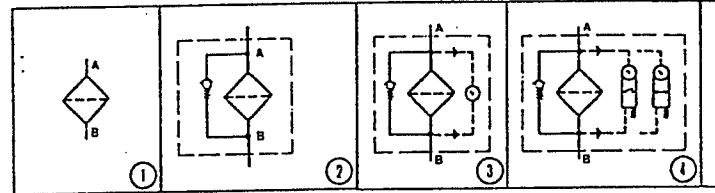
Supermicronic™ vst elements
with Δp 210 bar

Sm vst 3 $\beta_1 \geq 75$
Sm vst 10 $\beta_{10} \geq 75$
Sm vst 25 $\beta_{25} \geq 75$

at 16 bar differential pressure

Purolator Supermicronic™ filter elements are also resistant against HFA and HFC fluids.

6. Symbols



7. Part numbers pressure side installation

7.1 Housing design

7.2 Filter elements

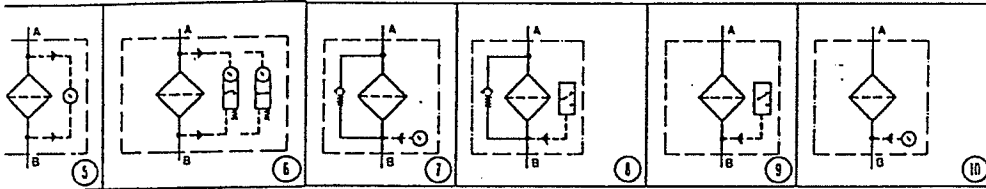
Part number	Type number	Nominal flow rate [l/min]	① Standard*	② With bypass valve	③ With bypass valve and visual indicator	④ With bypass valve and electrical indicator	⑤ With visual indicator	⑥ With electrical indicator	Supermicronic™ inorganic paper			
									Sm vst 3 Δp 210 bar	Sm vst 10 Δp 210 bar	Sm vst 25 Δp 210 bar	Sm 3 Δp 20 bar
		50							(470 cm²)	(470 cm²)	(470 cm²)	(590 cm²)
929 2205 100	Pi 2005-60								965 0509 153	965 0509 154	965 0509 155	
929 2205 101	Pi 2005-56											964 0509 15
929 2205 113	Pi 2005-57								[Pi 2205]	[Pi 3205]	[Pi 4205]	[Pi 2105]
929 2205 112	Pi 2005-58											964 0509 15
929 2205 103	Pi 2005-68								965 0509 153	965 0509 154	965 0509 155	
929 2205 102	Pi 2005-69								965 0509 153	965 0509 154	965 0509 155	
		80							(900 cm²)	(900 cm²)	(900 cm²)	(1150 cm²)
929 2206 100	Pi 2008-60								965 0510 153	965 0510 154	965 0510 155	
929 2206 101	Pi 2008-56											964 0510 15
929 2206 113	Pi 2008-57								[Pi 2208]	[Pi 3208]	[Pi 4208]	[Pi 2108]
929 2206 112	Pi 2008-58											964 0510 15
929 2206 103	Pi 2008-68								965 0510 153	965 0510 154	965 0510 155	
929 2206 102	Pi 2008-69								965 0510 153	965 0510 154	965 0510 155	
		150							(2010 cm²)	(2010 cm²)	(2010 cm²)	(2350 cm²)
929 2208 100	Pi 2015-60								965 0512 153	965 0512 154	965 0512 155	
929 2208 101	Pi 2015-56											964 0512 15
929 2208 113	Pi 2015-57								[Pi 2215]	[Pi 3215]	[Pi 4215]	[Pi 2115]
929 2208 112	Pi 2015-58											964 0512 15
929 2208 103	Pi 2015-68								965 0512 153	965 0512 154	965 0512 155	
929 2208 102	Pi 2015-69								965 0512 153	965 0512 154	965 0512 155	
		300							(3800 cm²)	(3800 cm²)	(3800 cm²)	(4420 cm²)
929 2209 100	Pi 2030-60								965 0513 153	965 0513 154	965 0513 155	
929 2209 101	Pi 2030-56											964 0513 15
929 2209 113	Pi 2030-57								[Pi 2230]	[Pi 3230]	[Pi 4230]	[Pi 2130]
929 2209 112	Pi 2030-58											964 0513 15
929 2209 103	Pi 2030-68								965 0513 153	965 0513 154	965 0513 155	
929 2209 102	Pi 2030-69								965 0513 153	965 0513 154	965 0513 155	
		450							(5600 cm²)	(5600 cm²)	(5600 cm²)	(6540 cm²)
929 2210 100	Pi 2045-60								965 0514 153	965 0514 154	965 0514 155	
929 2210 101	Pi 2045-56											964 0514 15
929 2210 113	Pi 2045-57								[Pi 2245]	[Pi 3245]	[Pi 4245]	[Pi 2145]
929 2210 112	Pi 2045-58											964 0514 15
929 2210 103	Pi 2045-68								965 0514 153	965 0514 154	965 0514 155	
929 2210 102	Pi 2045-69								965 0514 153	965 0514 154	965 0514 155	

5. Test regulations

Purolator filter elements are manufactured respectively, tested in accordance with the following international standards:

Designation	
ISO 2941	Hydraulic filter elements: Verification of burst resistance
ISO 2942	Hydraulic filter elements: Determination of fabrication integrity
ISO 2943	Hydraulic filter elements: Verification of material compatibility with hydraulic fluids
ISO 3723	Hydraulic filter elements: Method for testing end-cap load
ISO 3724	Hydraulic filter elements: Verification of flow fatigue characteristics
ISO 3968.2	Hydraulic filter elements: Evaluation of pressure drop versus flow
ISO 4572	Hydraulic filter elements: Testing of filter performance (Multi-Pass-Test)

The low pressure filter Pi 200 is in compliance with the German pressure vessel regulations 27.02.80.
The series has passed the required tests and is controlled according to the instructions of these regulations.

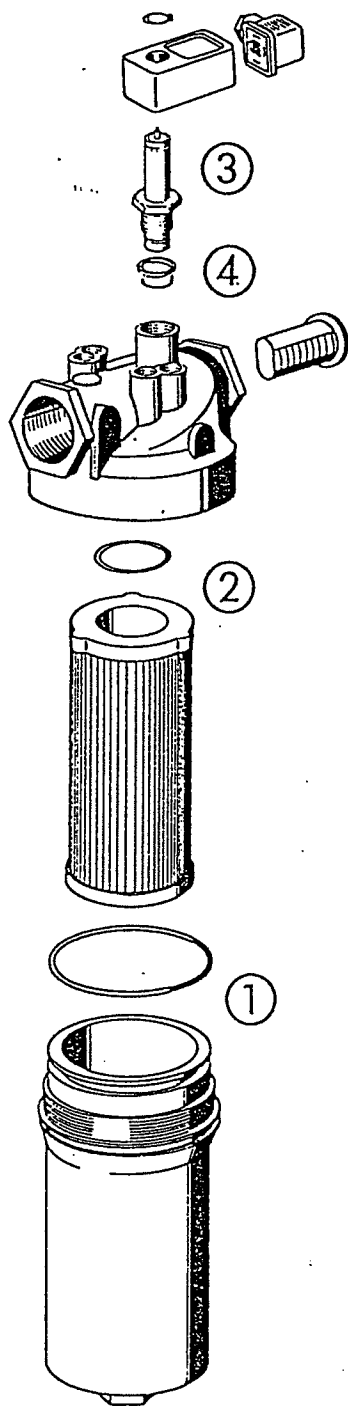


8. Part numbers suction side installation

8.1 Housing design

() = filter surface area [] = type number

Micronic* organic paper				⑩	⑨	⑧	⑦	②	①	Nominal flow rate [l/min]	Type number	Part number
Sm 10 Δp 20 bar	Sm 25 Δp 20 bar	Mic 10 Δp 20 bar	Mic 25 Δp 20 bar	With vacuum meter	With vacuum switch	With bypass valve 0.25 bars and vac. switch	With bypass valve 0.25 bars and vac. meter	With bypass valve	Standard*			
(590 cm ²)	(590 cm ²)	(640 cm ²)	(640 cm ²)							10	Pi 2005-60	929 2205 100
964 0509 151	964 0509 152	964 0509 103	964 0509 105								Pi 2005-67	929 2205 188
[Pi 3105]	[Pi 4105]	[Pi 1105]	[Pi 1005]								Pi 2005-62	929 2205 189
964 0509 151	964 0509 152	964 0509 103	964 0509 105								Pi 2005-61	929 2205 190
											Pi 2005-65	929 2205 182
											Pi 2005-66	929 2205 105
(1150 cm ²)	(1150 cm ²)	(1250 cm ²)	(1250 cm ²)							16	Pi 2008-60	929 2206 100
964 0510 151	964 0510 152	964 0510 103	964 0510 105								Pi 2008-67	929 2206 188
[Pi 3108]	[Pi 4108]	[Pi 1108]	[Pi 1008]								Pi 2008-62	929 2206 189
964 0510 151	964 0510 152	964 0510 103	964 0510 105								Pi 2008-61	929 2206 190
											Pi 2008-65	929 2206 182
											Pi 2008-66	929 2206 105
(2350 cm ²)	(2350 cm ²)	(2500 cm ²)	(2500 cm ²)							30	Pi 2015-60	929 2208 100
964 0512 151	964 0512 152	964 0512 103	964 0512 105								Pi 2015-67	929 2208 188
[Pi 3115]	[Pi 4115]	[Pi 1115]	[Pi 1015]								Pi 2015-62	929 2208 189
964 0512 151	964 0512 152	964 0512 103	964 0512 105								Pi 2015-61	929 2208 190
											Pi 2015-65	929 2208 182
											Pi 2015-66	929 2208 105
(4420 cm ²)	(4420 cm ²)	(4670 cm ²)	(4670 cm ²)							60	Pi 2030-60	929 2209 100
964 0513 151	964 0513 152	964 0513 103	964 0513 105								Pi 2030-67	929 2209 188
[Pi 3130]	[Pi 4130]	[Pi 1130]	[Pi 1030]								Pi 2030-62	929 2209 189
964 0513 151	964 0513 152	964 0513 103	964 0513 105								Pi 2030-61	929 2209 190
											Pi 2030-65	929 2209 182
											Pi 2030-66	929 2209 105
(6540 cm ²)	(6540 cm ²)	(6920 cm ²)	(6920 cm ²)							90	Pi 2045-60	929 2210 100
964 0514 151	964 0514 152	964 0514 103	964 0514 105								Pi 2045-67	929 2210 188
[Pi 3145]	[Pi 4145]	[Pi 1145]	[Pi 1045]								Pi 2045-62	929 2210 189
964 0514 151	964 0514 152	964 0514 103	964 0514 105								Pi 2045-61	929 2210 190
											Pi 2045-65	929 2210 182
											Pi 2045-66	929 2210 105



11. Installation, operating and maintenance instructions

11.1 Filter installation

When installing the filter make sure that sufficient space is available to remove filter element and filter bowl.

Preferable the filter should be installed with the filter bowl pointing downwards.

The contamination indicator must be visible.

11.2 Connecting the electrical contamination indicator

The electrical indicator is connected via a 2-pole appliance plug according to DIN 43 650 with poles marked 1 and 2.

The electrical section can be inverted to change from Normally Open position to Normally Closed position or visa versa.

11.3 When must the filter element be replaced?

- Filters equipped with visual and electrical contamination indicator:
During cold starts, the indicator may give a warning signal. Depress the red button of the visual indicator once again only after operating temperature has been reached. If the red button immediately pops out again and/or the electrical signal has not switched off after reaching operating temperature, the filter element must be replaced after the end of the shift.
- Filters without contamination indicator: the filter element should be replaced after the trial run or flushing of the system. Afterwards follow instructions of the manufacturer.
- Please always ensure that you have original Purolator replacement elements in stock: disposable elements (Micronic® or Supermicronic®) cannot be cleaned.

11.4 Element replacement

- Stop system and relieve filter from pressure.
- Unscrew the filter bowl by turning counter-clockwise. Clean the bowl using a suitable cleaning solvent such as pure benzine or kerosene.
- Remove filter element with a side-to-side motion.
- Check O-ring and back-up ring on the filter bowl for damage. Replace, if necessary.
- Make sure that the part number on the spare element corresponds with the part number of the filter label. Open the plastic bag and push element over the receiving piece in the filter head. Now remove plastic bag.
- Complete installation by screwing on the bowl, turning clockwise until it comes to a full stop. Back off the bowl 1/8 to 1/2 turn.

Subject to technical alteration without prior notice.

12. Spare parts list

		part no./housing	
		Pi 2005-2008	Pi 2015-2045
Pos. 1-2	Seal kit		Seal kit
	Buna	851 0475 020	Buna 851 0476 020
	Viton	851 0475 022	Viton 851 0476 022
	EPR	851 0475 024	EPR 851 0476 024
Pos. 3	visual	940 2191 103	electrical 940 2191 102
	Pis 3098		Pis 3097
			electrical upper part only 851 9530 000
Pos. 4	Seal kit for contamination indicator		
	Buna	851 0633 020	
	Viton	851 0633 022	
	EPR	851 0633 024	

Purolator Filter GmbH

Unternehmensbereich Industriefilter

Postfach 1309 · Schleibachweg 45

D-7110 Öhringen · Telefon (079 41) 671 · Telex 7 4 458

9. Specifications

Design:	line mounting filter
Operating pressure:	25 bars*
Test pressure:	60 bars
Temperature range:	-40 °C to +120 °C
Bypass opening pressure:	Δp 3,5 bars $\pm$ 10%
for pressure operation	Δp 0,25 bars $\pm$ 10%
for suction operation	
Filter head material:	GDAL
Filter bowl material:	AL/St
Material of seals:	Buna/AL
Activating pressure of visual/electrical differential pressure indicator:	Δp 2,2 bars $\pm$ 10%
Vacuummeter	-1 bar to 0 bar
Vacuumswitch	2000 mm WC
Electrical data of contamination indicator:	
Maximum voltage:	220 V~/=
Maximum current on contact:	1 A
Maximum contact load:	30 VA/20 W
Inrush current:	70 VA
Type of protection:	IP 65 when inserted and secured
Act:	bistable
Electrical connection:	PG 11 $\varnothing$ 6-10

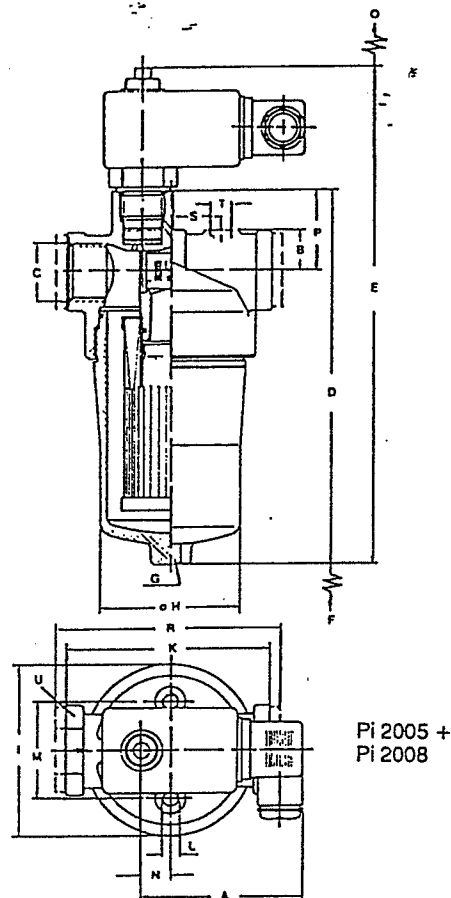
If desired, electrical indicator may be supplied with a light. The electrical indicator function can be changed from the Normally Open position to the Normally Closed position or visa versa by inverting the electrical section.

With the inrush current of 70 VA the indicator can trigger small contactors or contactor relays.

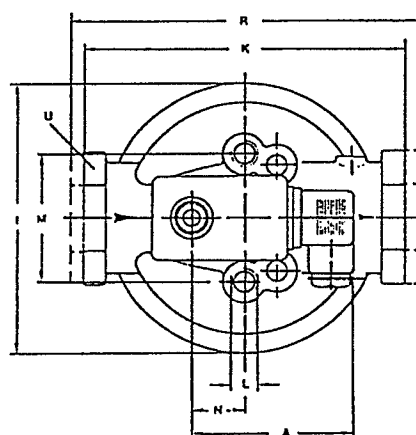
Inductivity in the direct current may require the use of a signal eraser.

For further information also see our leaflet: Contamination indicators.

* Types Pi 2005 and 2008 have an operating pressure of 60 bar.



Pi 2005 +
Pi 2008



Pi 2015 -
Pi 2045

10. Dimensions

Dimensions Type	A	B	C	D	E	F	G	H	I	K	L	M	N	O	P	R	S*	T*	U	Weight
Pi 2005	78	19	G½	172	225	80	19	66	80	-	109	45	13	M 8 x 10	36	45	23,5	G½	36	0,9 kg
Pi 2008	78	19	G¾	248	301	80	19	66	80	95	-	45	13	M 8 x 10	36	45	23,5	G½	36	1 kg
Pi 2015	78	30	G1	244	297	110	24	109	128	-	164	60	24,5	M 12 x 15	55	45	40	G½	55	2,1 kg
Pi 2030	78	30	G1¼	360	413	110	24	109	128	150	-	60	24,5	M 12 x 15	55	45	40	G½	55	2,4 kg
Pi 2045	78	30	G1¼	475	528	110	24	109	128	150	-	60	24,5	M 12 x 15	55	45	40	G½	55	6,5 kg

* only by suction side installation

Dial thermometer

Class 1.5
Series 8201

in metal case
for panel mounting or self-supporting
Protection IP53 or IP65 for TZ 047
Case size: $\varnothing$ 60mm, $\varnothing$ 80mm and $\varnothing$ 100mm
Bezel: 72 x 72mm, 96 x 96mm

Brief description

The dial thermometer has a liquid-filled or gas-filled measuring system and is used for temperature measurement e.g. in HVAC, refrigerating plants, photographic processing equipment, washing machines, on cookers and grills, in machinery and mechanical equipment, as well as in cooling and tank lorries.

Features

- case, bezel ring, bezel or flange in anti-corrosion stainless steel
- gas-filled for environmental and health protection
- Protection IP53 or IP65 with TZ 047 (extra Code)
- indication adjustment at the rear
- vibration-resistant through damping liquid with TZ 009 (extra Code)

Type designation

8201 Series 8201
Dial thermometer in metal case

Standard accessory

Operating Instructions B 80.1100

Case sizes

8201-.-60 case 60mm dia.
(not for Style 02)
8201-.-80 case 80mm dia.
8201-.-10 case 100mm dia.
8201-25-72 bezel 72 x 72mm
8201-24-96 bezel 96 x 96mm

Ordering examples

Dial thermometer

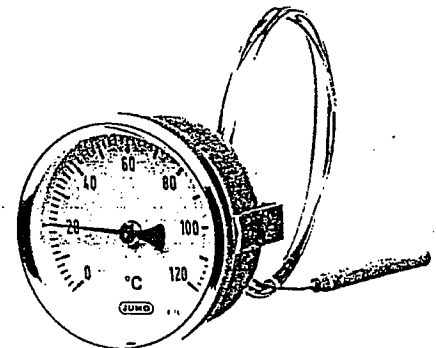
Type: 8201-10-80
Scale: 0 to +200°C
Extra Code: TZ 002-006
Temperature probe: TF 01, copper, 6mm dia.
Probe mounting: TA 03 1/2" pipe, brass,
S = 150mm

Dial thermometer

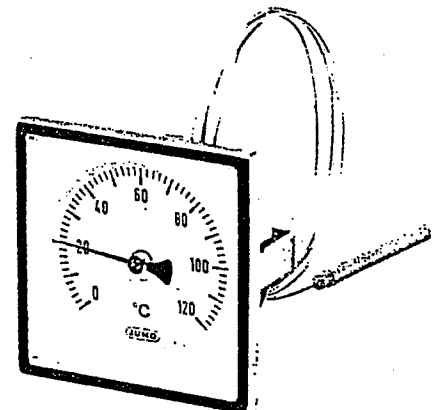
Type: 8201-23-10
Scale: 0 to +120°C
Extra Code: TZ 018
Capillary: FL 02, 2000mm
Temperature probe: TF 01, copper, 6mm dia.
Probe mounting: TA 22, 1/2" pipe A, brass,
S = 200mm

Style

8201-02	
8201-10	
8201-20	
8201-21	
8201-22	
8201-23	
8201-24	
8201-25	



Type 8201-21-60



Type 8201-25-72

Ranges (AB)

(similar to DIN 16 203)

Scale °C	Measuring range °C	Accuracy °C
-40 + 40	- 30 + 30	1.5
-30 + 50	- 20 + 40	1.5
-20 + 120	0 + 100	3.0
0 + 60	+ 10 + 50	1.5
0 + 80	+ 10 + 70	1.5
0 + 100	+ 10 + 90	1.5
0 + 120	+ 20 + 100	3.0
0 + 160	+ 20 + 140	3.0
0 + 200	+ 20 + 180	3.0
0 + 250	+ 30 + 220	4.0
+50 + 250*	+ 70 + 230	3.0
0 + 300*	+ 30 + 270	6.0
+50 + 300*	+ 80 + 270	4.0
0 + 350*	+ 50 + 300	6.0
+50 + 350*	+ 80 + 320	6.0
0 + 400*	+ 50 + 350	6.0
0 + 450*	+ 50 + 400	6.0
0 + 500*	+ 50 + 450	8.0
0 + 600*	+ 100 + 500	10.0

* ranges at extra charge

other scale ranges on request

Technical data

(details for standard version)

Case

stainless steel, Mat. Ref. 1.4301

Bezel ring, bezel or flange

stainless steel, Mat. Ref. 1.4301

Scale

white, black figuring, other versions on request

Window

glass, PMMA (Plexiglass) for Styles 24 and 25, TZ 009 and TZ 047

Indication

linear, pointer movement $260^\circ \pm$ approx.

Measuring system

liquid-filled for ranges up to 350°C
gas-filled for ranges 400°C upwards

Indication accuracy

see accuracy in table of ranges (AB)

Response time

90 % response of liquid-filled measuring system approx. 20sec, measured in agitated water bath with 6mm dia. probe.

90 % response of gas-filled measuring system approx. 5sec, measured in agitated oil bath with 8mm dia. probe

Ambient temperature error in % of scale range

referred to deviation from $+23^\circ\text{C}$ reference value

- on case
 - liquid-filled measuring system 0.15% of span per $^\circ\text{C}$ change in ambient temperature;
 - on gas-filled measuring system 0.05% of span per $^\circ\text{C}$ change in ambient temperature
- on capillary (per mm)
 - liquid-filled measuring system 0.015% of span per $^\circ\text{C}$ change in ambient temperature;
 - gas-filled system no error;
 - increased ambient temperature results in higher temperature reading.

Indication adjustment

In the event of ambient temperatures which are constant but differ considerably from the reference value $+23^\circ\text{C}$, the error can be corrected by rotating the slotted bolt on the back of the case. Not possible with TZ 009 and TZ 047 (extra Codes)

Nominal position

NL90, symbol $\perp$

Limit temperatures for storage and transport

-30 to $+70^\circ\text{C}$

for range -40 to $+40^\circ\text{C}$ up to 50°C max.

for range -30 to $+50^\circ\text{C}$ up to 60°C max.

Subject to modification.

Capillaries (FL)

see Data Sheet 80.8003, page 1 and 2, Table 2

Temperature probes (TF)

see Data Sheet 80.8003, page 1 and 2, Table 1

Probe mountings (TA)

see Data Sheet 80.8003, page 3, Table 3
(TA20, 23 and 25 are not available on Styles 02 and 10)

Pockets (SH)

see Data Sheet 80.8003, page 4, Table 4

Extra Codes (TZ)

TZ 002 peak-reading pointer, on case 60, 80 and 100mm dia. (not available with TZ 009 or TZ 047)

TZ 004 adjustable marker, on case 60, 80 and 100mm dia. (not available with TZ 009 or TZ 047)

TZ 006 case with bimetal compensation

TZ 009 case with damping liquid is available for the following types:

8201-20-60
8201-22-60
8201-23-60
8201-20-10
8201-22-10
8201-23-10

TZ 018 metal bezel ring, bezel or flange, black

TZ 028 capillary reinforcement at case and probe (not with FL 21 and FL 22)

TZ 047 Protection IP65 and EN 60529, is available for the following types:

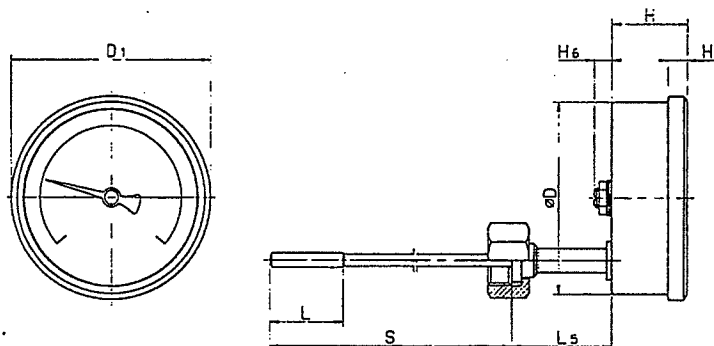
8201-20-60
8201-22-60
8201-23-60
8201-20-10
8201-22-10
8201-23-10

TZ 060 scale to customer specification

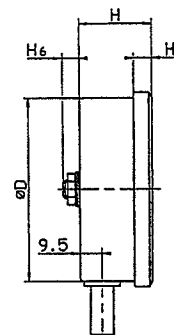
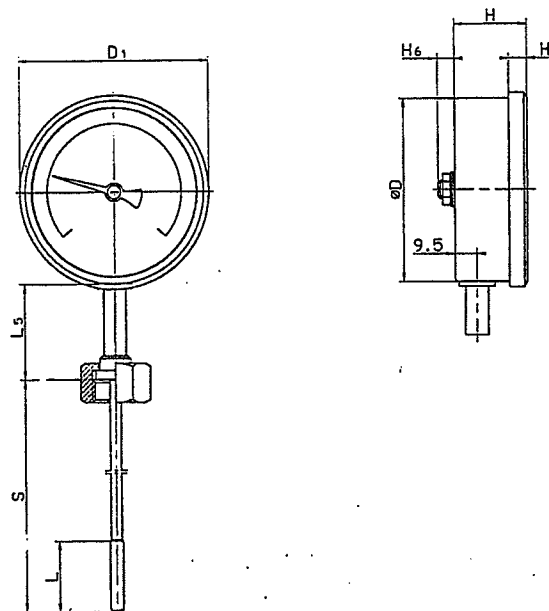
TZ 086 peak-reading pointer adjustable by screwdriver; protected by screw-on cap (not available with TZ 009 or TZ 047)

Dimensions

Types: 8201-02-80
8201-02-10



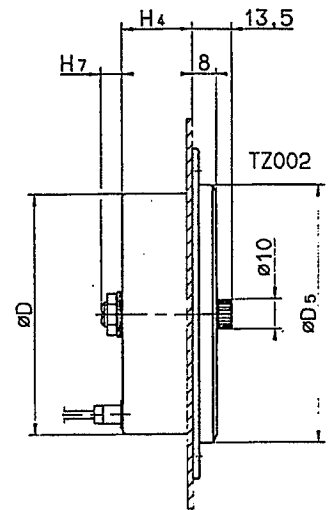
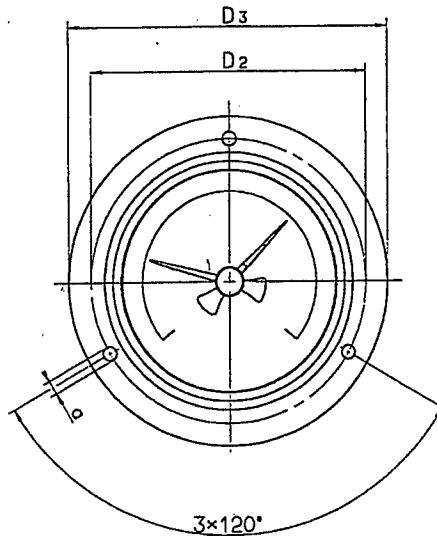
Types: 8201-10-60
8201-10-80
8201-10-10



Dimensions

Types: 8201-20-60
8201-20-80
8201-20-10

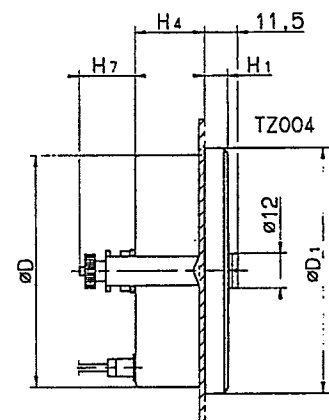
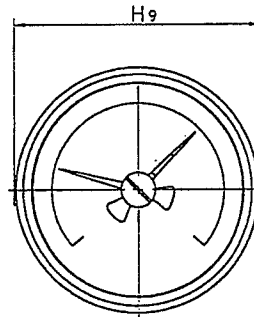
Panel cut-out
for case 60mm dia. = $62^{+0.5}_0$ mm dia.
for case 80mm dia. = $82^{+0.5}_0$ mm dia.
for case 100mm dia. = $102^{+0.5}_0$ mm dia.



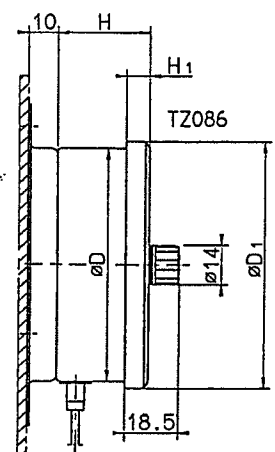
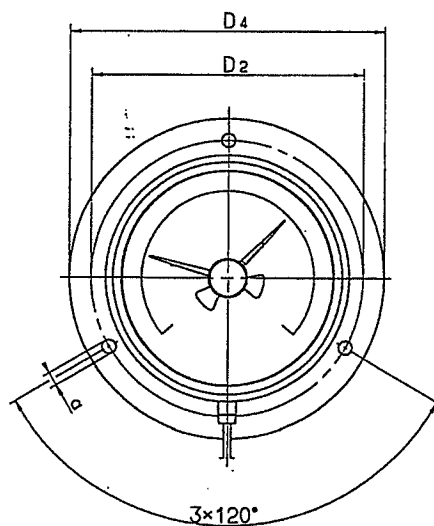
Types: 8201-21-60
8201-21-80
8201-21-10

Panel cut-out
for case 60mm dia. = $62^{+0.5}_0$ mm dia.
for case 80mm dia. = $82^{+0.5}_0$ mm dia.
for case 100mm dia. = $102^{+0.5}_0$ mm dia.

width of mounting bracket

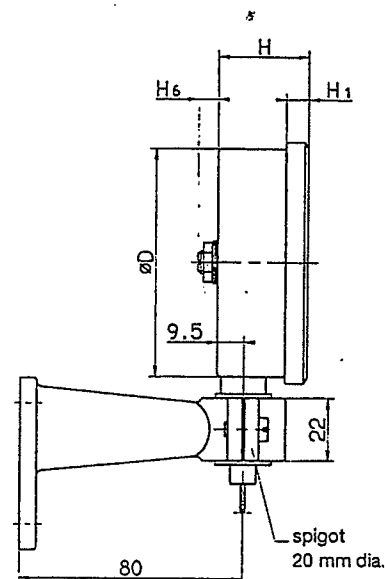
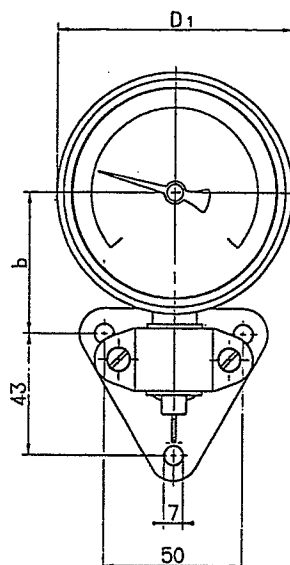


Types: 8201-22-60
8201-22-80
8201-22-10



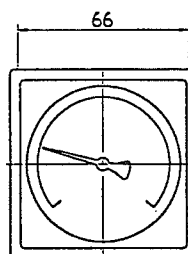
Dimensions

Types: 8201-23-60
8201-23-80
8201-23-10

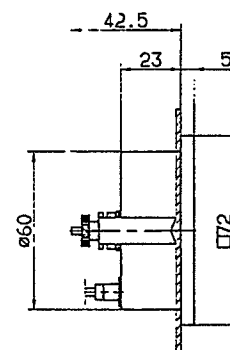


Types: 8201-25-72

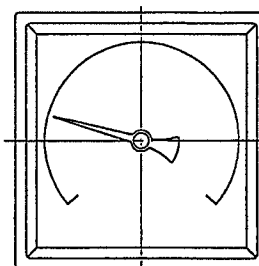
width of mounting bracket



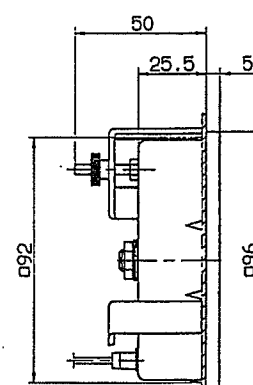
Panel cut-out: $62^{+0.5}_0$ mm dia.



Type: 8201-24-96



Panel cut-out: $92 \times 92^{+0.8}_0$ mm



Case dia.	H	H ₁	H ₄	H ₆	H ₇	H ₉	D	D ₁	D ₂	D ₃	D ₄	D ₅	a	b	L ₅
60	30 (36)	7	23 (28)	approx. 7.5	19.5	66	60	65	75	85	86	66	3.6	39.5 (50)	41.5*
80	32.5	8	24.5			86	80	85	95	110	110	86	4.8	49.5	
100	33 (36)		25 (28)			106	100	106	116	132	132	107.5		59.5 (70)	

* on mounting TA 02, L₅ = 70.5mm max.
on mounting TA 22 and TA 31, L₅ = 49.5mm
values in brackets apply to TZ 009 or TZ 047
Subject to modification.

Float Switches

23

Type-form

Float switch

Sliding tube P= PVC
M= yellow brass
E= high-grade steel

Number of contacts

A - upper contact
B - lower contact
C - auxiliary contact

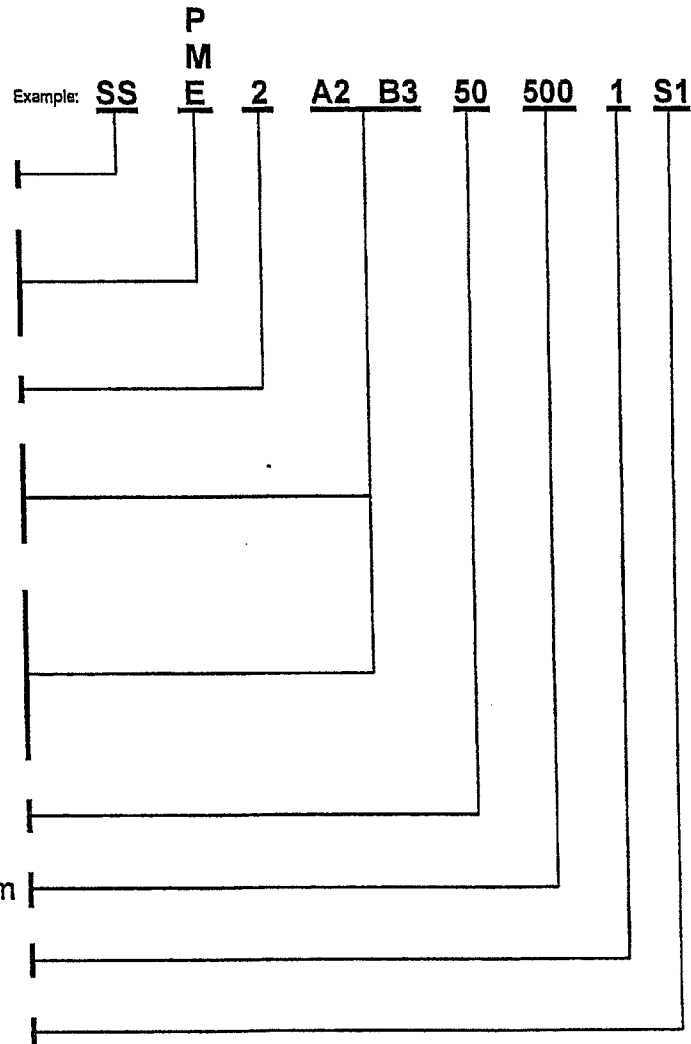
1 - closes if level rises
2 - opens if level rises
3 - closes if level decreases
4 - opens if level decreases
5 - change-over contact

50 -breaking-point-length L1 e.g. 50mm

500 -breaking-point-length L2 e.g.500mm

1 - design number

S1 - float e.g. 35x40mm



ZENGLER

Specification:

Switching power max.:

normally open contact: 250 VA
normally closed contact: 250 VA
change over contact: 60 VA

Switching voltage max.:

normally open contact: 250 V
normally closed contact: 250 V
change over contact: 220 V

Switching current max.:

normally open contact: 3 A
normally closed contact: 3 A
change over contact: 1 A

Repeatable accuracy:

level-difference 5 mm

Max. contacts / switch:

4 normally open contacts
4 normally closed contacts or
3 change over contacts

Connecting housing made of:

Al-die casting (colour grey) or
Polyamid

Protection:

IP 65

Cable-entry:

PG 11

cable-connection:

1,5 mm²

operating pressure:

operating temperature: on request

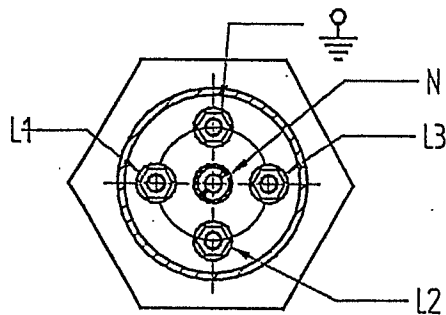
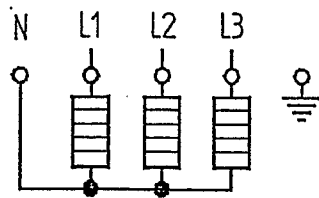
Temperature switch:

Switching voltage max: 220V

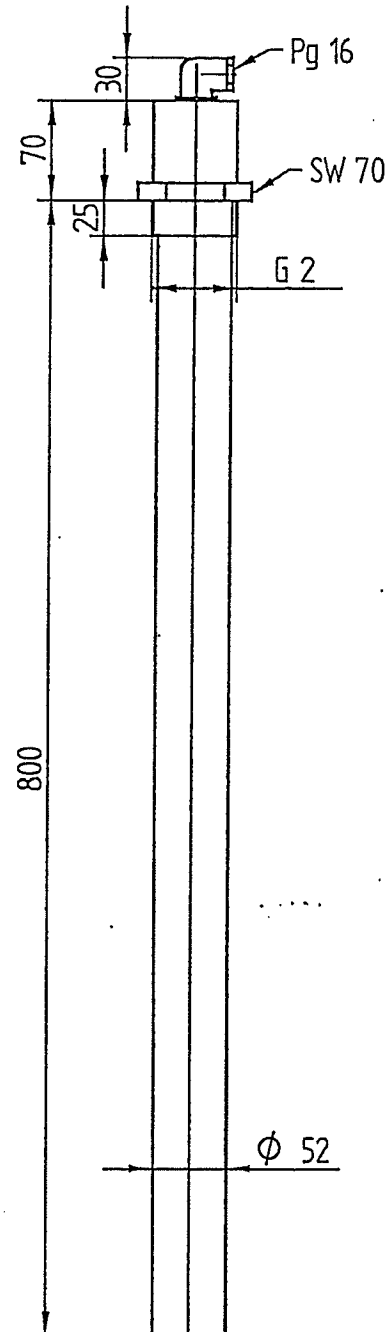
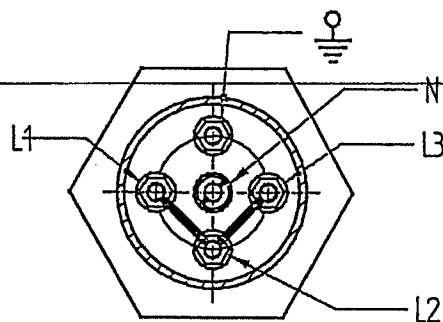
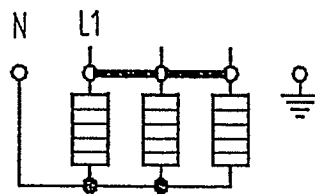
Switching current max: 2A ..

C. C. O. 2

Schaltbild 3 x 440 V
circuit diagram 3 x 400 V



Schaltbild
circuit diagram 230 V



3 x 440 V	1090 W
3 x 400 V	900 W
230 V	900 W

Typ: O / 52
Best.-Nr. 001-1305-009-385
Ref-No.
Hersteller Helios
Supplier

28.

Type: WBV 10.

Order-No.: 001-1305-009-568

Threaded thermoresistance element

Measuring element: Pt 100 according to DIN 43760

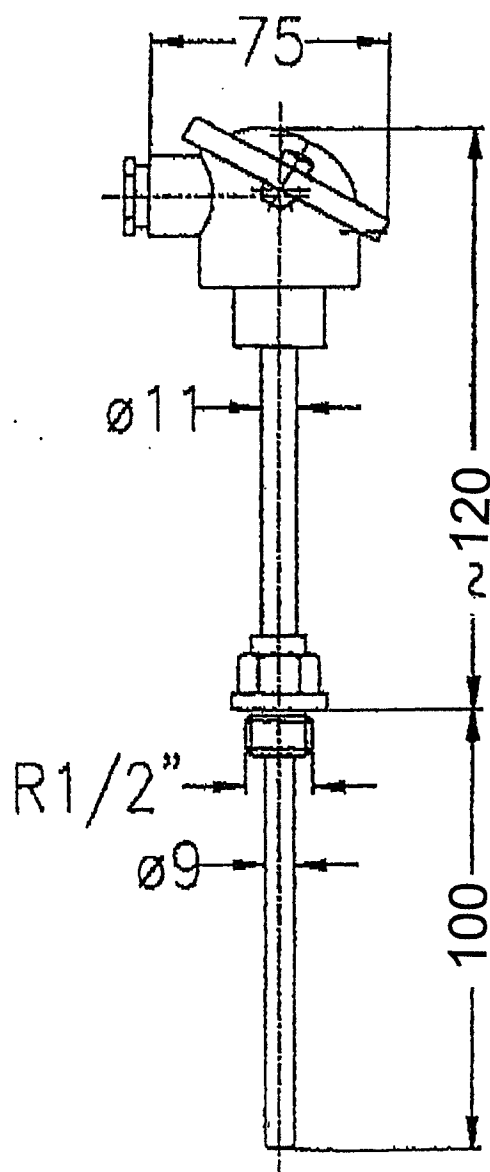
Screwed male: R 1/2"

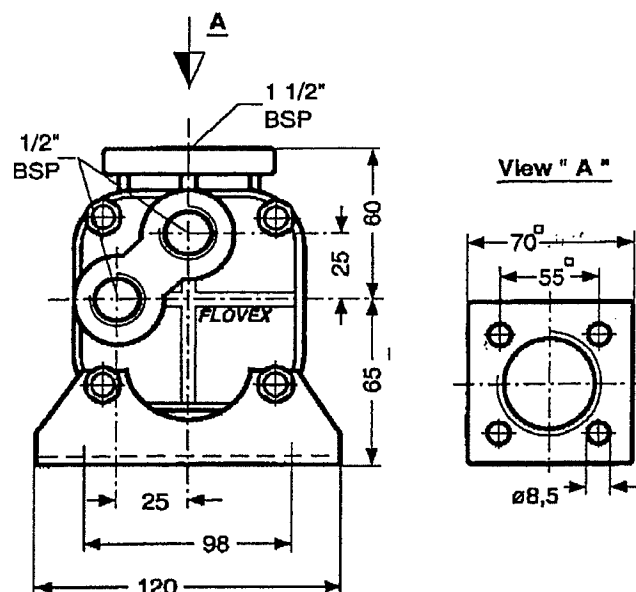
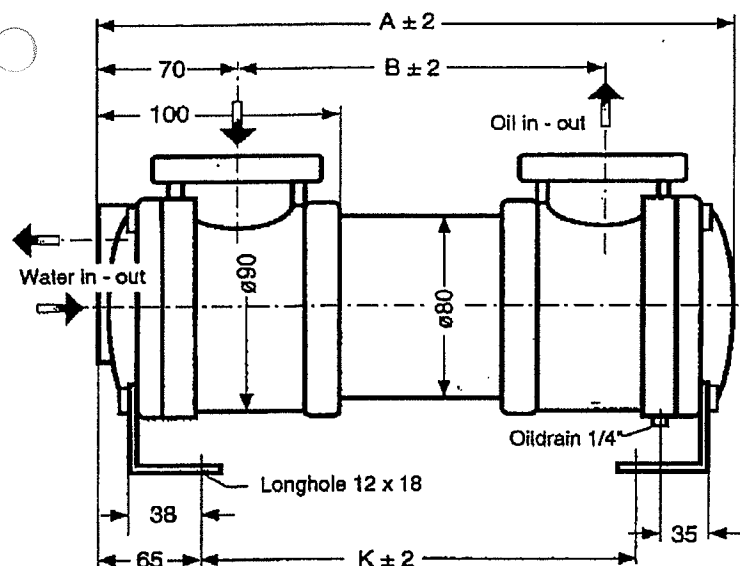
Terminal head: light alloy, B24x1,5

Protection tube: stainless steel X10 CrNiMoTi 1810 material no. 1.4571

Max. temperature: 400°C

Weight: 0,4 kg





TYPE	A	B	K	Weight
T 8035	425	285	305	4,5
T 8061	680	540	560	6,5
T 8076	830	692	712	7,5
T 8091	985	845	865	8,0
T 8106	1140	998	1018	11,0
T 8120	1290	1152	1172	12,0

	Shell side	Tube side
Operating pressure	16 bar	10 bar
Test pressure	24 bar	15 bar
Bursting pressure	95 bar	36 bar
Bursting pressure (tube)	--	105 bar
Operating temperature	100 °C.	100 °C.
Test temperature	150 °C.	150 °C.

Connections

Oil connection standard with 1 1/2" and 70 mm square flange. The oil and water connections can be chosen freely. The counterflow system remains due to the 4-way design.

General

The series T 80 is the result of a continuous adjustment to the demand for efficient small heat exchangers with low water consumption. In addition to this, a heat exchanger is required which eliminates the problems during cleaning of the bundle tubes.

- 4-way system on the water side
- low water consumption
- stainless steel bundle head 1.4435
- sea water design T-OT
- tight diversion baffles on the oil side
- fin tube with 1,25mm wall thickness
- clear tube diameter 10,5 mm
- tubes rolled into the bundle head
- cleaning without dismantling on the water side
- cleaning on the water side during works of plant
- compensates heat tensions in the housing

The installation of these heat exchanger should be made vertically with the water connection at the bottom. This mainly applies for the installation of a continual control in the water inlet. The additional installation of a water filter prevents sedimentation completely.

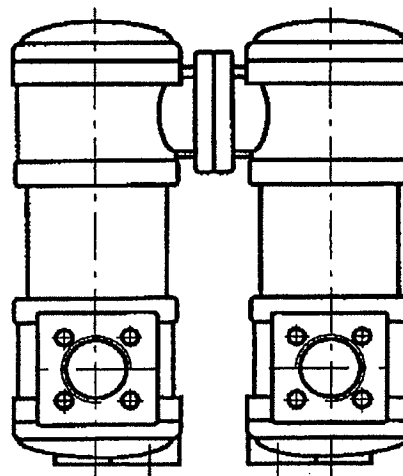
The oil and water connections can be chosen freely on the inlet and outlet side and can be adjusted to space demand. The counterflow system will remain due to the 4-way system on the water side.

Material

Housing	Aluminium
Cover	Radlion-A-RV- 50, fiber-glass reinforced
Cover thread	Cadmium-plated, the threads 1/2" are a continous steel insert
Bundle head	Aluminium diecasting stainless steel 1.4435 plated
O-rings	Oil- and water side in Nitril
Bundle	T = SF-Cu-normal water T/OT = CuZn20Al2 for bracklish- and sea-water with zink anodes in the cover

Design

This series can be supplied with connections in reverse order, by 90°, in every direction. If the oil is very stringy, with small quantities it might be possible to use a double heat exchanger, in order to prevent pockets in the diversions. The oil connection of the heat exchangers can thus be made via the



flange without an additional tube connection. The position of the connections must be stated when the order is placed. Attention! Heat exchangers with connection heads in reverse order of the series T 80 have a delivery time of 2-3 weeks. For vertical installation you can use our console K.01.0102.

Performances

The tables with the performance data for selection of the cooler can be seen on sheet 11.A. The values for conversion are mentioned on sheet 02.A "Example for Calculation" in this catalog.

Installation

If possible and the space is sufficient, the installation of the heat exchangers should be made vertically. Thus possible impurities will be washed out in the bundle even with small water quantities.

The sedimentation of impurities can thus be prevented to a great extent. Furthermore all bundle tubes are equally filled with water. The use of a water filter prevents sedimentation of impurities completely.

Special information has to be observed, if the cooler is being operated with well water. We do not mention this here, as well water is rarely used these days.

- In case of well water please contact us!

The coolers of series T 80 have a very low weight because they are made of Aluminium. The coolers mounted with flanges on the oil side, can easily be attached to the pipeline. Even in case of small tank surfaces, it is not necessary to look for installation space. The cooler can practically be hung next to the tank.

- The water and oil connections can be chosen freely because of the double counterflow system, and they can be adjusted to the pipelines and the space demand.
- The series T 80 compensates length extensions in case of very high temperatures in the housing.

Remarks

In order to obtain a cost saving and good use of the cooler capacity, the water filtration and the kind of the water regulation must be adjusted to each case.

This should therefore be considered, if 2- or 3-point regulation should be applied. The material of the bundle tubes is a priority for this decision.

- Special kinds of brass of the CuNi group have to be observed specially, as there is a danger of localized corrosion at water speeds up to 0,9 m/s. This range can only be excluded by an electric 2-point regulation. When using such OPEN-CLOSED regulation, however, the above maximum admissible flow speed must be taken into consideration.
- Because of the bundle material of the FLOVEX coolers, the undesired range in the water speed of 0 - 0,9 m/s can be disregarded, and the inexpensive 3-point regulation can be applied without auxiliary energy. The top limitation will be made automatically, and the low speed does not matter with the material used by us. FLOVEX coolers were designed for a water speed of maximum 2,7 m/s. The limit to be observed is safely at 6,0 m/s.

Inspection and Cleaning

Due to the kind of the bundle, the inside diameter of the bundle tubes and the entire construction of the cooler, the inspection and the cleaning are very simple. The bundle can be inspected after taking off the rear cover, i. e. after loosening some screws and closing of the water inlet.

The bundle tubes have an inside diameter of 10,5 mm. These tubes can easily be cleaned if necessary. It would be best to put a 10 mm thread bar through the tubes, if necessary with light blows of the hammer.

The possible lime layer will come off immediately because the surface of the inner side of the tubes is smooth.

- A cleaning with chemical agents is not possible

The fin tubes with a wall thickness of 1,25 mm (measured without fin) are very solid.

Due to the large inside diameter of 10,5 mm of the bundle tubes, larger impurity particles do not disturb the operation, as they are normally washed out.

Contrary to the heat exchangers with 4 + 6 mm tubes, the use of dirt

taps is not necessary for tubes with a large inside diameter.

If a chemical cleaning is necessary, this should only be made as follows:

- Clean the cooler approx. 2 hours with 5% HCl in the water, wash and soak it in a closed basin (do not use in a closed rooms).
- Then the cooler should be washed for 2 hours with 5 % NaOH in the water for neutralisation of the acid used before.
- This procedure is very extensive and must be made very carefully so that no aggressive remainders stay in the bundle tubes.

With our system of large-dimensioned tubes, a mechanical cleaning with thread bars or a brush is sufficient. The heat exchanger does not even have to be dismantled.

If the dismantling of the cooler becomes necessary, this can be made without removing the pipeline.

By simple loosening of the cover on the water side, the flanges on the oil side or screws, as well as the fixing screws at the consoles, the entire cooler can be pulled aside. The space required for this, beside or behind the cooler, is identical with the basic space of the cooler.

For systems, which have been heavily stressed by impurities, we supply special design with bundle tubes of 14 mm Ø. (In this coherence please notice the design according T.E.M.A.).

These heat exchangers are often being used for heat recovery from tall oil baths. These coolers have free access on both tube ends without those tubes disturbing during cleaning and without having to dismantle connections for the cleaning procedure (for example the series AEW).

Guarantee

In the frame of our "General conditions for sale and delivery" we assume no guarantee for collisions damages or impurities, falling out and defects by reason of wrong maintenance and installation spaces. The time of guarantee is 12 months after erection, but not longer than 18 months after dispatch of the heat exchanger.